

VASCULAR SURGERY

CHAPTER

1

Acute Limb Ischemia

GENERAL CONSIDERATIONS

Arteries are characterized by the presence of a collateral circulation, which under normal conditions is collapsed. This circulation opens whenever the blood flow in the main artery is inadequate for the tissue requirements.

Definitions

Ischemia

- The term *ischemia* means ‘relative diminished blood supply in relation to tissue demand’. Ischemia may be *acute* or *chronic* according to the onset of arterial occlusion.

Acute Limb Ischemia

- Acute limb ischemia is any *sudden decrease or worsening in limb perfusion* causing a potential threat to extremity viability. It is a surgical emergency condition, which requires timely management and can be associated with high morbidity and mortality rates

Effects of Ischemia Depend on

1. *Type of artery*: Some arteries have a very efficient collateral circulation to the extent that their ligation *may not* be followed by serious consequences e.g. subclavian artery. On the other hand, the popliteal artery has a poor collateral circulation.
2. *Rate of artery occlusion*: Acute ischemia is much more serious than chronic ischemia as there is not enough time for the collateral circulation to develop.
3. *State of the collateral vessels*: Healthy collaterals can compensate to some extent the ill-effects of ischemia. If collaterals are atherosclerosed, ischemia will be severe and extensive.
4. *General condition of the patient*: Myocardial insufficiency or severe anemia will exacerbate the effects of ischemia.

ETIOLOGY

1. *Acute arterial embolism* (of a relatively healthy arterial tree).
2. *Acute arterial thrombosis* (of a relatively diseased arterial tree).
3. *Acute traumatic ischemia*:
 - Arterial *trauma* either, direct (stab) or indirect (blunt trauma or fracture bones).
 - Trauma may cause:
 - a) Blood vessel injury (sub-intimal hematoma or thrombosis of the lumen).
 - b) Spasm of the vessel.
 - c) Complete transection of the vessel.
4. *Other causes*
 - Accidental intra-arterial drug injection (Figure 1).
 - Aortic dissection.
 - External compression, e.g. a very tight tourniquet.
 - Compartment syndrome: increased pressure due to hematoma or tissue edema in a closed space by bone, inter-osseous membrane and deep fascia resulting in ischemia (more in tibial plateau fracture).
 - Extensive ilio-femoral deep vein thrombosis (DVT) (e.g. phlegmasia cerulea dolens)



Figure 1: Acute ischemia causing gangrene of the fingers of the left hand due to accidental intra-arterial drug injection.

- The most common causes for non-traumatic acute limb ischemia include:
 1. Embolism from the heart.
 2. Thrombosis over an existing arterial plaque/stenosis.
 3. Thrombo-embolism from an arterial aneurysm.
 4. Acute bypass graft occlusion (vein or synthetic graft).

ARTERIAL EMBOLISM

Definition

- Embolus is originally a *Greek* word that means "*a plug or a stopper*".
- Embolism is the passage of a matter from one part of the circulation to another through a vascular lumen. *Arterial embolism is considered the classical example of acute ischemia.*

Etiology

1. Thrombus

- The usual source of arterial emboli is a *thrombus* present, either in the left side of the heart, or in a major proximal artery.

2. Cardiac Sources

- *Arrhythmia*: Atrial fibrillation (AF) is usually associated with rheumatic mitral stenosis (MS) or with atherosclerosis that is complicated by a thrombus of *left atrium*. The latter is the source of emboli.
- *Recent myocardial infarction* (MI) that causes a mural thrombus.
- *Bacterial endocarditis* on top of rheumatic or congenital heart disease, or affecting a prosthetic valve.

3. Non-cardiac Sources

- A thrombus in an aneurysm e.g.
 - a) Abdominal aortic aneurysms (AAA).
 - b) Cervical rib causes post-stenotic dilatation (subclavian aneurysm), which results in distal embolization.
- Platelet thrombi on top of an ulcerated atherosclerotic plaque, e.g. in the carotid arteries that → cerebral embolization → transient ischemic attack and even hemiplegia.

4. Iatrogenic

- Mitral valvotomy.
- Valve replacement.
- After arterial puncture in arteriogram.
- In therapeutic embolization to treat bleeding (e.g. peptic ulcer).

Pathophysiology of Arterial Embolization

- It involves the following steps (Figure 2).
 1. Impaction of the embolus.
 2. Propagation.
 3. Muscle necrosis.
 4. Gangrene – venous thrombosis

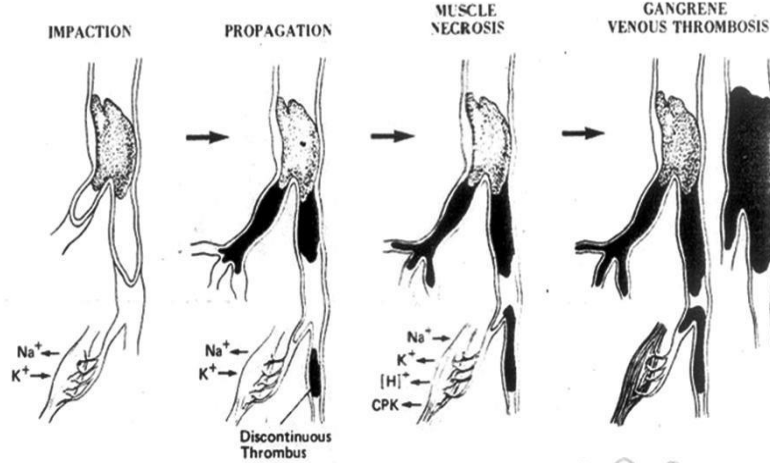


Figure 2: Patho-physiology of arterial embolization

Sites of Emboli

- An embolus in the arterial tree is usually arrested at the sites of *bifurcation of arteries* where sudden decrease in the size of the arterial lumen occurs (Figure 3). The lower limbs (LLs) are more commonly affected

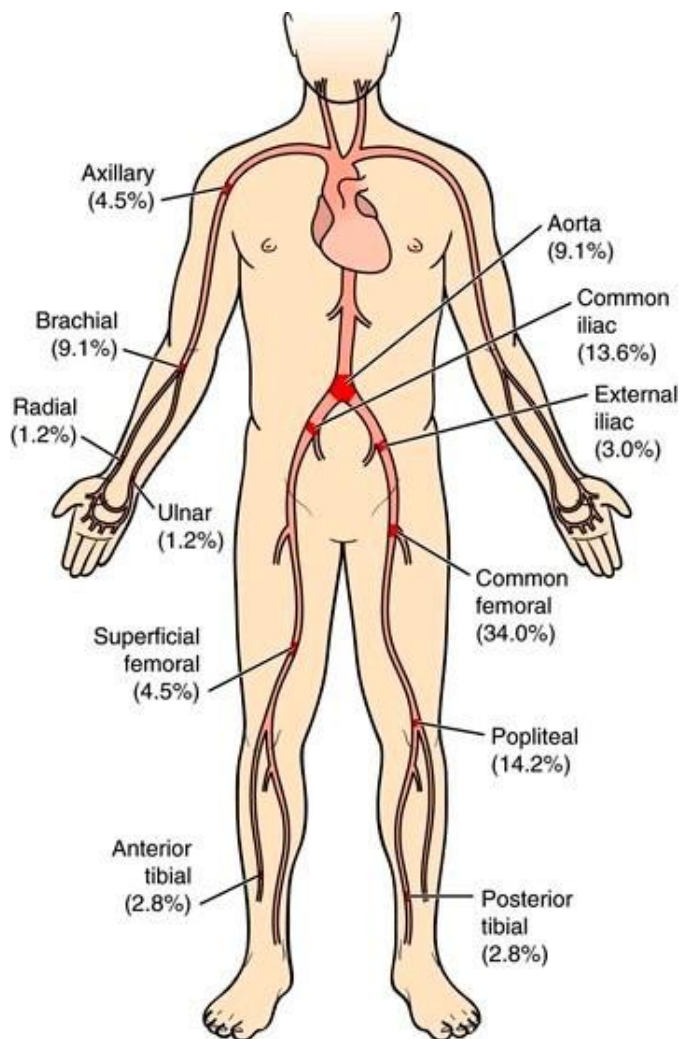


Figure 3: Common sites of arrest of arterial emboli (sites of bifurcation of arteries).

- The commonest sites of arrest of emboli are:
 1. The bifurcation of the common femoral into superficial & deep femoral arteries (40%).
 2. Aortic bifurcation (*saddle embolus*) (Figure 4).
 3. Bifurcation of the popliteal artery.
 4. Bifurcation of the brachial artery.
 5. Bifurcation of common carotid artery (CCA)

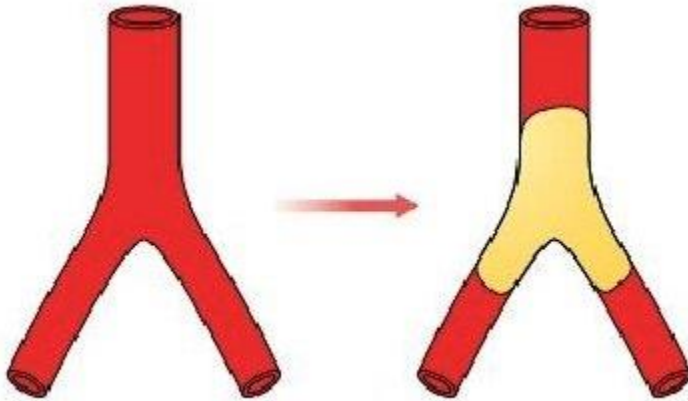


Figure 4. Embolism at aortic bifurcation (*saddle embolus*). The source may be atrial fibrillation or recent myocardial infarction

ACUTE ARTERIAL THROMBOSIS

Definition

Sudden occlusion of an artery by the formation of an adherent thrombus over a *diseased intima* e.g. ulcerated atherosclerosis.

Etiology

1. On top of chronic occlusive arterial disease eg atherosclerosis (commonest) or Buerger's disease
2. As a complication of arterial aneurysm.
3. As a result of traumatic contusion of vessels.
4. Arterial thrombosis may complicate febrile illness or gastroenteritis especially in children. Dehydration → hemo-concentration → thrombosis, heart failure and hypertension.
5. Blood disease that increases blood coagulation.
6. Drugs that increase blood coagulation e.g. corticosteroids, oral contraceptives.
7. Debilitating diseases.

Factors Predisposing to Acute Thrombosis

There are certain predisposing factors for development of thrombosis in a normal or diseased artery (Table 1).

Table 1: Predisposing factors for acute arterial thrombosis

Predisposing factors for development of acute arterial thrombosis	
Cause	Example
Dehydration	Hot weather, diarrhea, fever
Hypotension	Shock, sepsis, Myocardial infarction
Pressure	Prolonged sitting
Malignancy	Hematologic cancers
Hypercoagulable state	Thrombophilia

Pathological Consequences of Acute Ischemia

- After circulatory arrest, widespread distal intravascular thrombosis occurs. Thrombosis may also affect the artery proximal to the site of occlusion.
- *Ischemic muscles* get swollen and this, in turn, exaggerates the effects of ischemia by compression of the collaterals (*compartment syndrome*). Early fasciotomy can avoid this problem (Figure 5) i.e. longitudinal incision of the deep fascia of the leg in the affected compartment [anterior, lateral, posterior (superficial and deep)], either *open* or *semi-closed* fasciotomy.



Figure 5. Fasciotomy to avoid compartment syndrome

- If involving all compartments, ***fibulectomy*** is done.
- Consequences depend upon the etiology, site and duration of ischemia as well as the efficiency of treatment (Figure 6):
 1. Complete recovery (in small embolus with efficient collaterals).
 2. Gangrene, which is usually of the *wet type*.
 3. Chronic ischemia.
 4. Volkmann's ischemic contracture.

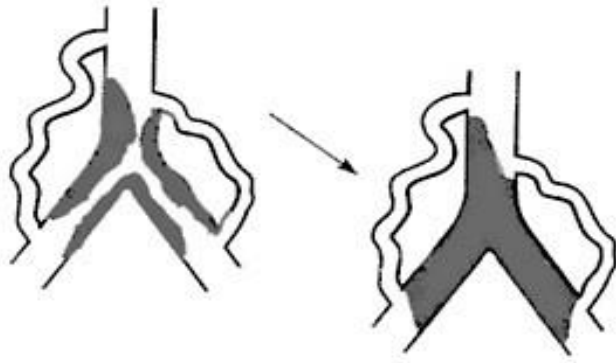


Figure 6: Consequences of vascular occlusion

Clinical Features of Acute Ischemia

Classic Symptoms and Signs (6Ps) of Severe Acute Ischemia

1. Pain
2. Paresthesia
3. Pulseless
4. Pallor (color) (symptom)
5. Paralysis or paresis
6. Perishing coldness (temperature)

Inspection (Color)

- Initially the leg is marble *white* and the veins are empty (Figure 7). After 6-12 hours, vasodilatation occurs and capillaries fill with stagnant deoxygenated blood leading to *cyanosis and mottled appearance*.
- Later, capillaries rupture causing fixed mottling and cyanosis (Figure 8), "irreversible changes", with tense calf, fixed plantar flexion (*rigor mortis*) and then, frank *gangrene*



Figure 7: Early changes: Pallor of the foot with empty veins,



Figure 8: Late changes: Fixed mottling and cyanosis (arrow),

Palpation

Pulseless

- Palpate arterial pulses (Figure 9). Compare with the other side.
- Slow capillary re-filling of the skin after finger pressure.

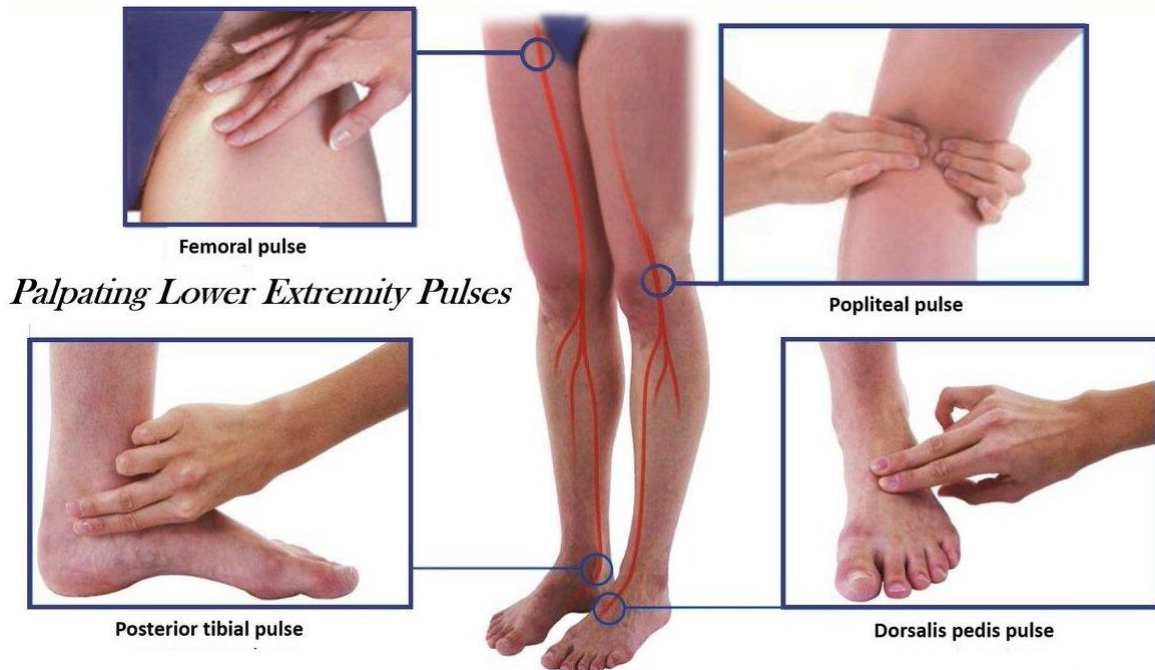


Figure 9. Palpation of arterial pulses of the lower limb (femoral, popliteal, posterior tibial and dorsalis pedis)

Temperature

- The limb is cold with a level of temperature change.

Parasthesia (loss of sensory function) occurs first

- Numbness will progress to anesthesia.
- The progress of sensory loss is in that order: light touch, vibration sense, proprioception, deep pain and pressure sense (late).

Paralysis / paresis (loss of motor function)

- Loss of motor function indicates advanced limb-threatening ischemia.
- Later → irreversible ischemia → muscle turgidity.
- Intrinsic foot muscles are affected first, followed by leg muscles.
- Detecting early muscle weakness is difficult because toe movements are produced mainly by leg muscles.

Rutherford Classification

Rutherford classification of acute ischemia is summarized in Table 2.

Table 2: Rutherford Classification of Acute Ischemia

Rutherford Clinical Categories of Acute Limb Ischemia					
Category	Description/prognosis	Findings		Doppler signals	
		Sensory loss	Muscle weakness	Arterial	Venous
I. Viable	Not immediately threatened	None	None	audible	audible
II. Threatened;					
a. Marginally	Salvageable if promptly treated	Minimal (toes) or none	None	(Often) inaudible	Audible
b. Immediately	Salvageable with immediate revascularization	More than toes, associated with rest pain	Mild, moderate	(Usually) inaudible	Audible
III. Irreversible*	Major tissue loss or permanent nerve damage inevitable	Profound, anesthetic	Profound, paralysis (rigor)	Inaudible	Inaudible

Anterior Tibial Compartment Syndrome

Definition

- Increased pressure due to hematoma or tissue edema in a closed space by bone, interosseus membrane & deep fascia resulting in ischemia (more in tibial plateau fracture) or in delayed cases of acute ischemia.

Clinical Picture

- At first, the pulse & skin circulation are normal, but the early manifestations are:
 1. Severe pain not proportional to the trauma.
 2. Marked muscle tenderness.
 3. Passive stretch of muscles causes severe pain.
 4. Loss of sensation in the 1st dorsal cleft of foot.

Treatment of Compartmental Syndrome

- Elevation of the leg
- Drugs to reduce edema.
- If progressive → *fasciotomy* of the affected compartment [anterior, lateral, posterior (superficial & deep)], either *open*, or *semi-closed* fasciotomy. If involving the 4 compartments, *fibulectomy* is performed

Investigations for Acute Ischemia

The aims of investigation are:

1. To diagnose the condition (which is usually diagnosed on clinical basis)
2. To define the etiology of the condition.
3. To evaluate co-morbidities of the patient that will have mostly a surgical intervention.

Imaging Studies

1. *Duplex scan* localizes & identifies the presence of embolism or a thrombus.
2. *Arteriography*. It is *not* done in a threatened limb. Its value is in cases of diagnosed acute thrombosis because it provides information that is essential before doing an arterial reconstruction. This information includes:
 - Site of occlusion
 - Proximal inflow, i.e. if there is another proximal arterial narrowing.
 - Flow distal to the occlusion, i.e. distal run-off.
3. *Echocardiography* detects cardiac sources of embolism.
4. *Plain X-ray chest* (cervical rib, aortic aneurysm), ECG, ECHO.
5. *Plain X-ray bone* (trauma)

Laboratory Studies

1. Elevated blood urea nitrogen (BUN) and creatinine indicate intra-vascular hypovolemia due to fluid sequestration in the limb.
2. Acidosis and elevated CPK and WBCs indicate extensive muscle necrosis.

Differential Diagnosis

1. Low flow states (shock).
2. Acute occlusion of popliteal or femoral aneurysm.
3. Phlegmasia cerulea dolens (PCD) (edema, cyanosis and tense calf muscle)
4. Aortic dissection.
5. Arterial spasm after trauma (ischemia improves by vasodilators)

Differences between Embolism and acute thrombosis on top of atherosclerosis are summarized in Table 3. Differences between embolic and thrombotic arterial occlusion are shown in Table 4, and between acute ischemia and chronic ischemia in Table 5.

Table 3: Differences between Embolism and acute thrombosis on top of atherosclerosis

Parameter	Embolism	Acute thrombosis on top of atherosclerosis
<i>History</i>	Arrhythmia or recent myocardial ischemia	Intermittent claudication
<i>Source of emboli</i>	Usually present e.g. cardiac	Absent
<i>Radial pulse</i>	Usually irregular (AF)	Usually regular
<i>Skin color</i>	White	Dusky
<i>Limb nutrition</i>	Normal	Picture of chronic ischemia
<i>Angiography</i>	- Sharp cut-off - Minimal collaterals	- Tapering stenosis - Extensive collaterals - Diffuse atherosclerosis

Table 4: Differences between embolism and acute arterial thrombosis

Parameter	Embolism	Thrombosis
- Duration of symptoms	2-4 hours (sudden onset)	24-48 hours (insidious onset)
- Symptom aggravation	Rapid	Slow
- Previous claudication	Absent	May be present
- Cardiac disease	May be present	May be present
- Contralateral pulses	Present	May be absent
- Multiple sites	Up to 15%	Rare
- Angiography	Total occlusion, no collaterals	Collaterals noted
- Artery on palpation	Soft, dark wall due to underlying clots	Hard, calcified
- Extracted clots	Dark red, gelatinous	White, firm
- Treatment	Embolectomy	Thrombolysis with angioplasty / bypass.

Table 5: Differences between acute and chronic ischemia

Parameter	Acute Ischemia	Chronic Ischemia
- Age	Young	Old
- History	Cardiac problems	Other manifestations of atherosclerosis
- Source of emboli	Present	Absent
- Preceding claudication	Absent	Present
- Motor function	Absent (no collaterals)	Present

Treatment of Acute Ischemia

Aim of Treatment

1. Prevent further propagation of the thrombus
2. Correct general condition of the patient
3. Restore blood supply as fast as possible to the limb

Algorithm of Management

An algorithm for management of acute limb ischemia is depicted in Figure 10

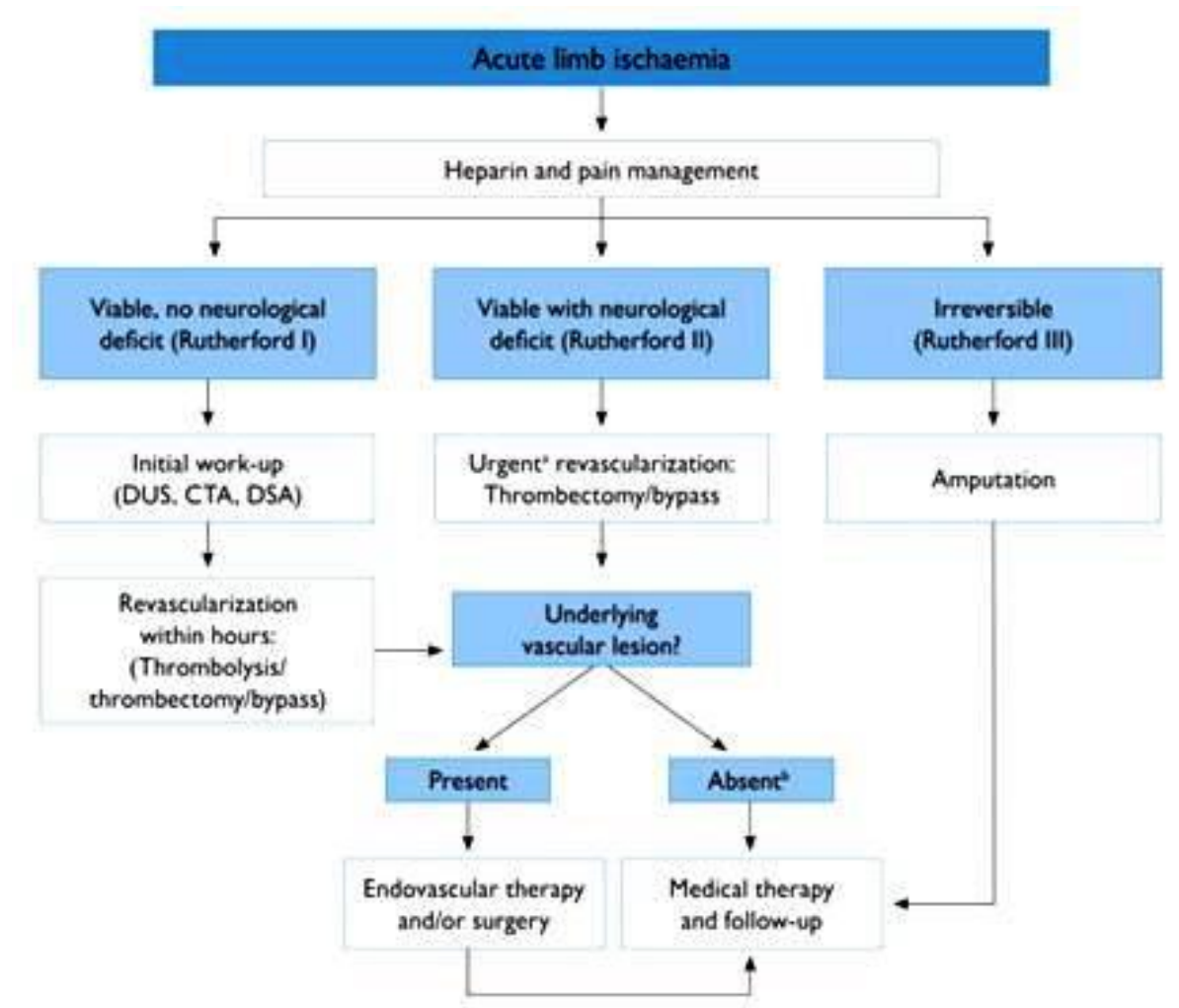


Figure 10. Algorithm for treatment of acute limb ischemia

Immediate Care

1. **Anti-coagulation:** IV Heparin. Start with 5000 to 10000 IU, followed by 3000-5000 IU/2 h. The dose is controlled by checking activated partial thromboplastin time (APTT) every 12 h, which should be maintained at 2-3 times the baseline level.
2. **Analgesics:** for pain.
3. **Measures to improve existing perfusion** + IV fluids to correct dehydration.
4. **Treatment of associated cardiac condition.**

Specific Measures

Treatment of Embolism

- *Emergency surgery is essential to remove the embolus “embolectomy”.* Urgent embolectomy using Fogarty balloon catheter is the standard method for removal of arterial emboli.
- The operation should be done as long as the limb is viable. The earlier the operation is done the better are the results because delay → occlusion of collaterals by propagated thrombosis & death of the limb.

- The operation can be done, under local, general or epidural anesthesia.
- *Femoral artery embolectomy* is done via a common femoral transverse arteriotomy.
- *Aortic bifurcation embolectomy* is done via bilateral femoral arteriotomy (Figure 11).

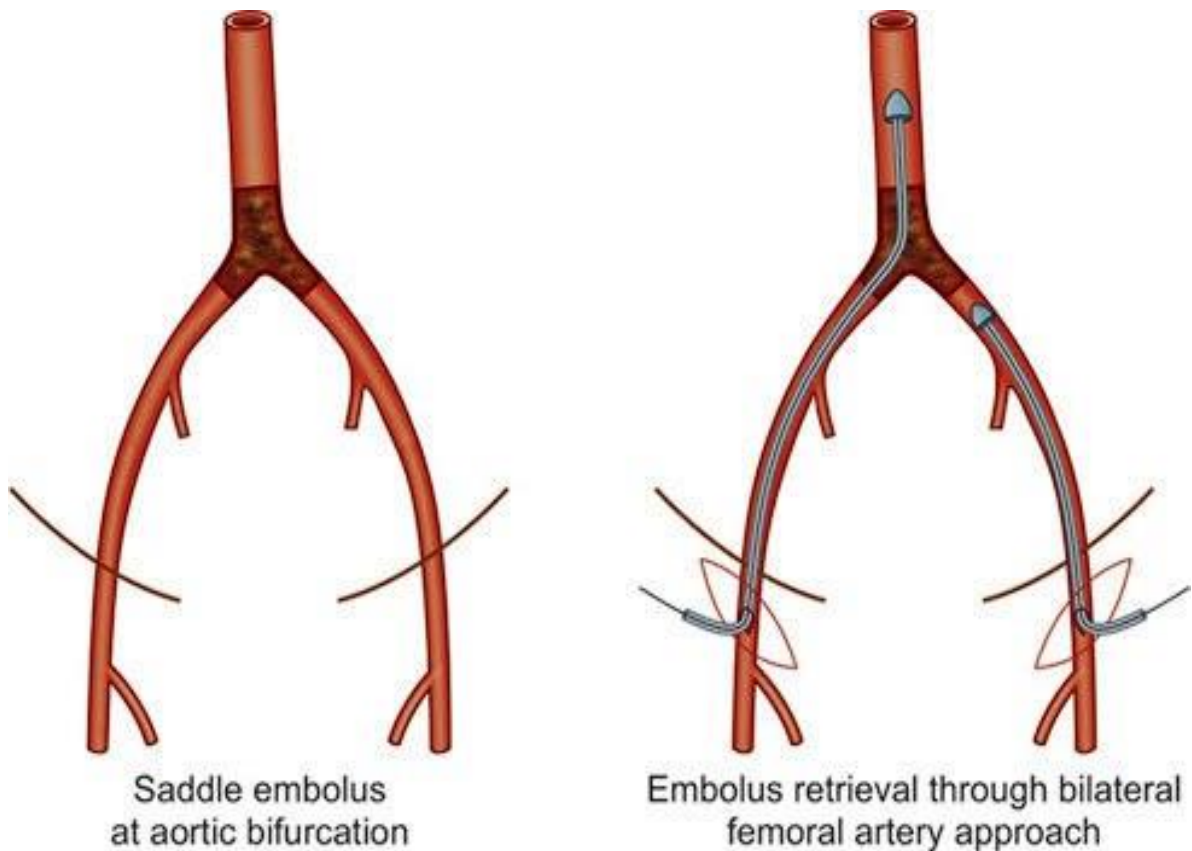


Figure 11. Aortic bifurcation embolectomy through bilateral femoral artery approach

- The Fogarty catheter is introduced through the femoral arteriotomy and is then passed up past the embolus in the lower aorta.
- The balloon is inflated and the catheter is pulled out. The balloon sweeps the embolus out of the arteriotomy. The strong arterial blood flow will clear any residual fragments.
- The Fogarty catheter should also be passed distally to extract any fragments of thrombi and propagated clots.
- *Brachial artery embolectomy* is done via a brachial artery exposure.
- *Arteriography* is done on table to ascertain clearance of the arterial tree.
- *Heparin* should be continued post-operatively until the cardiac condition is assessed and the need for further heparin therapy is determined.
- The *source of arterial emboli* should be corrected if possible.
- *Post-operative medical care:* When you do embolectomy after a long period of ischemia, you need to do *fasciotomy* as the limb will swell (post-perfusion).

Treatment of Acute Thrombosis

- Urgent arteriography is performed to plan for emergency “revascularization surgery”.
- If there is time for the use of “thrombolytic therapy”, elective revascularization surgery is done latter on.

1. Thrombolytic Therapy (CDT) Catheter Directed Thrombolysis

Thrombolytic drugs

- Streptokinase may cause anaphylaxis.
- Urokinase is a direct plasminogen activator.
- Tissue plasminogen activator (TPA)

Technique

- At the completion of arteriography the tip of the arterial catheter is embedded in the thrombus. The thrombolytic drug is gradually introduced by drip for 6-12 hours until the artery patency and limb viability are assured.
- Heparin and oral anti-coagulants are then continued.

Complications

- Anaphylactic reaction with the use of streptokinase.
- Hemorrhage at the puncture site, GIT or cerebral.

Indications of catheter-directed thrombolysis

- Viable or marginally-threatened limb (class I, IIa).
- Recent acute thrombosis (not suitable for embolism or old thrombi).
- Avoid patients with contra-indications

Contra-indications of catheter- directed thrombolysis

Absolute:

- Cerebro-vascular stroke within the previous 2 months.
- Active bleeding or recent gastrointestinal bleeding within the previous 10 days.
- Intra-cranial trauma or neuropathy within the previous 3 months.

Relative:

- Cardio-pulmonary resuscitation (CPR) within the previous 10 days
- Major surgery or trauma within the previous 10 days.
- Uncontrolled hypertension.

2. Revascularization procedures

- Thrombectomy
- Bypass procedures.

3. Treatment of Trauma

- The vessel is exposed.
- Any tension or bone fragment is removed.
- The bone is better fixed by external fixator and damaged segment of the artery is resected and replaced with a reversed saphenous graft.

Complications of Acute Limb Ischemia

These patients are at a high risk of myoglobinuria. The eventual requirement for amputation is also higher in these patients. Acute limb ischemia can lead to various loco-regional as well as systemic complications.

Systemic Complications

- Acidosis
- Hyperkalemia
- Myoglobinuria and acute tubular necrosis
- Cardiac arrhythmias
- Acute respiratory distress syndrome
- Septic shock

Local Complications

- Compartment syndrome
- Muscle necrosis
- Muscle infarct
- Causalgia (Reflex sympathetic dystrophy)
- Gangrene

1. GANGRENE

Definition

Gangrene means ‘macroscopic death of tissues caused by loss of blood supply’.

Etiology

1. *Ischemic*: Thrombosis – embolism - vasospastic
2. *Neuropathic*: Diabetes mellitus (DM)
3. *Traumatic* (physical / chemical): Burns – frost-bite
4. *Infective*
5. *Venous*: Phlegmasia cerulæ dolens (PCD)

Cardinal Signs of Gangrene

1. Loss of pulsations
2. Change of skin color
3. Loss of heat
4. Loss of sensation
5. Loss of function

Cardinal Types of Gangrene (Figure 12)

1. Dry gangrene
2. Wet gangrene
3. Gas gangrene

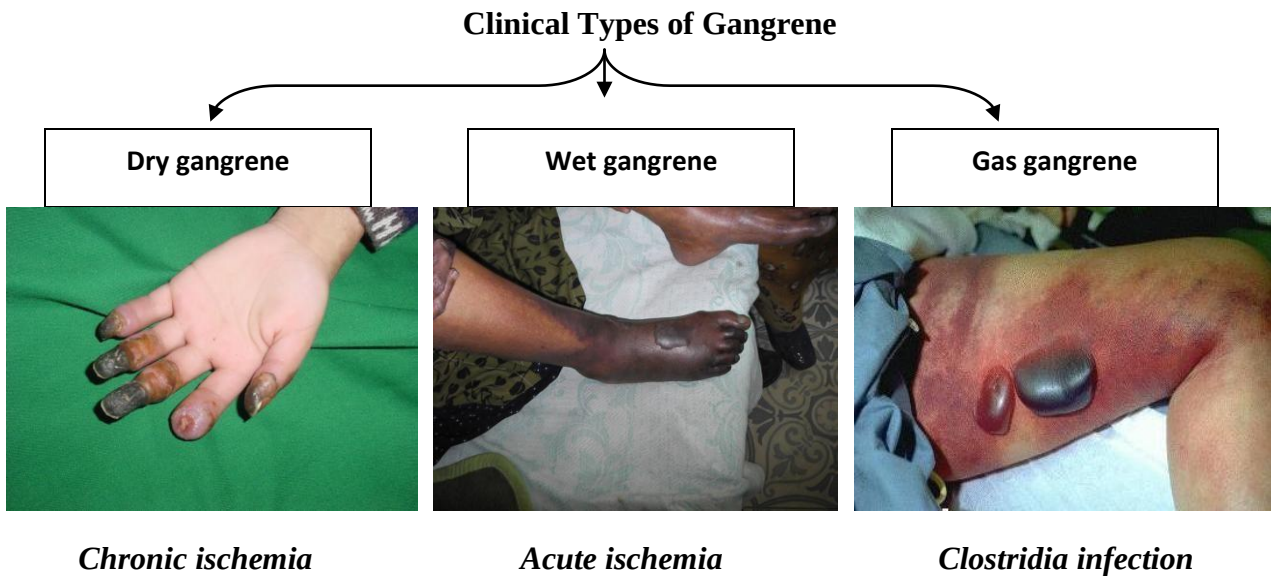


Figure 12: Clinical types of gangrene (dry gangrene – moist gangrene – gas gangrene)

Specific Types of Gangrene (Figure 13)

1. Necrotizing fasciitis
2. Fournier gangrene

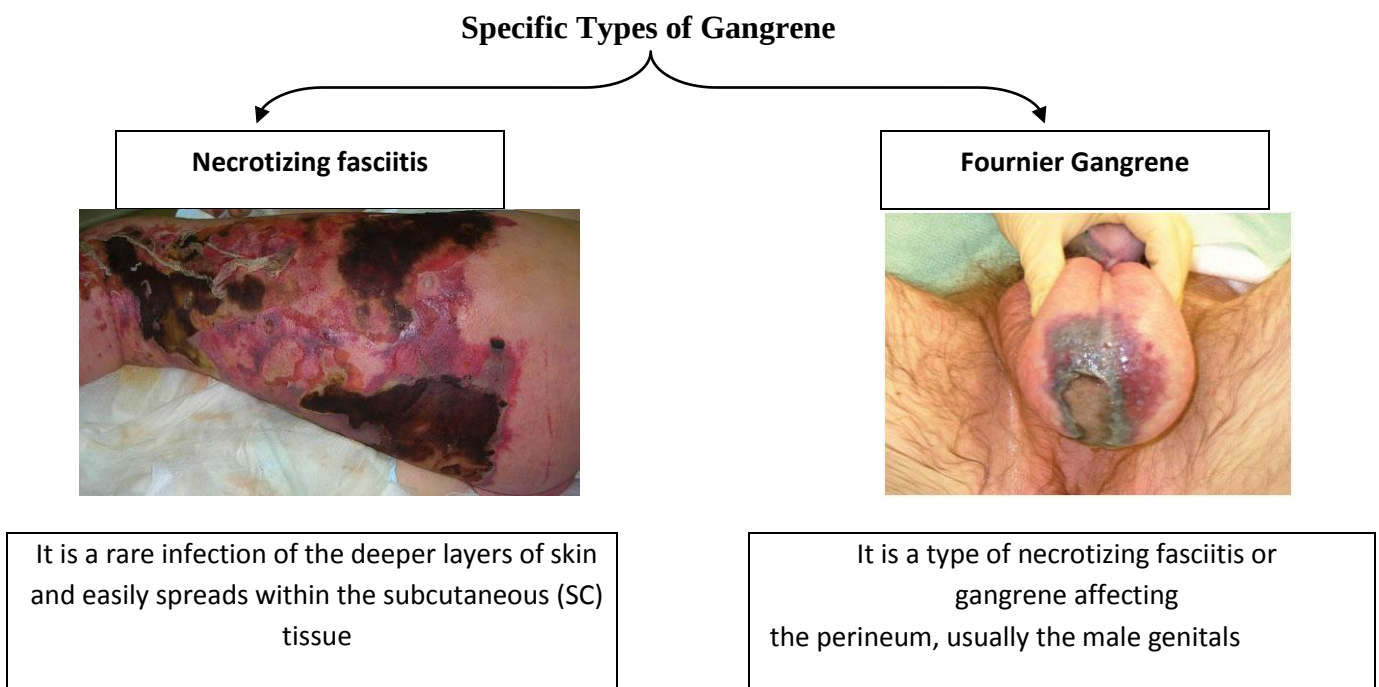


Figure 13: Specific types of gangrene (necrotizing fasciitis – Fournier gangrene)

AMPUTATION

Indications

1. Peripheral vascular disease
2. Diabetic limb disease
3. Necrotising fasciitis
4. Trauma (severe tissue damage) – traumatic amputation
5. Infection (chronic disabling infection, Gas gangrene) Tumours (malignant)
6. Nerve injury (trophic ulceration – insensitive limb)
7. Congenital anomalies (e.g. extra digits) Gross deformity.

Complication

1. Failure of healing
2. Infection
3. Phantom sensation: The patient feels that the amputated part is still present
4. Telescoping: It is the sensation that the distal part of the amputated extremity has moved proximally up the arm.
5. Phantom pain: The patient feels the amputated limb is painful. It disappears with time
6. TENS (trans cutaneous electric nerve stimulation)
7. Contractures
8. Neuroma

Criteria of Ideal Stump

- Optimum length (Table 6 below)

Level	Site of Amputation	Stump not less than
- Below knee	10-12 cm from tibial tuberosity	8 cm
- Above knee	12 cm above knee joint	20 cm from greater trochanter
- Upper Limb	Stump should be as long as possible.	

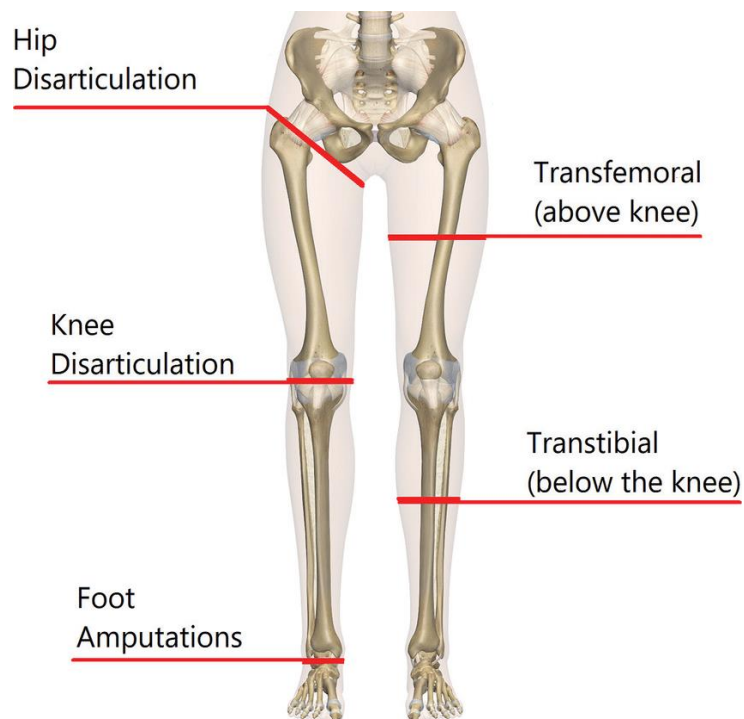
- Smooth and rounded
- Normal vascularity
- No projecting spur of bone
- The stump should not be under tension
- The scar should be away from pressure area
- Painless
- The stump is either; *side-bearing* (which is preferred) or end-bearing.

Levels of Amputation in the Lower Limb (Table 7 and 8)**Table 7:** Levels of distal amputations of the lower limb

Distal Amputation	
<i>Toe amputation</i>	In small vessel disease with good surrounding blood supply.
<i>Ray-excision</i>	Toe with part of the metatarsal bone
<i>Trans-metatarsal amputation</i>	Through the tarsal bones with long planter flap
Chopart amputation	Leave the <i>calcaneus</i> and <i>talus</i> bones. Disadvantages are muscle imbalance leading to equinus deformity.

Table 8: Levels of major amputations of the lower limb (Figure 14)

Major Amputation	
Syme's amputation	Remove all foot and the limb stump is left on tibia and fibula and covered with skin of the heel. It is <i>not</i> suitable for ischemia but only for trauma.
<i>Below-knee (BK)</i>	A long posterior flap is made of the gastrocnemius muscle or equal flaps.
<i>Above-knee (AK)</i>	With equal flaps.
<i>Through-knee</i>	Via the joint.
<i>Disarticulation</i>	The whole limb is removed
<i>Hind quarter</i>	For malignancy, the lower limb and part of iliac bone are excised.

**Figure 14.** Levels of major amputations of the lower limb

CHAPTER

2

Chronic Ischemia

Chronic ischemia means ‘inadequate blood supply to tissue’. This is a slowly progressive arterial obstruction that gives enough time for collaterals to develop and, therefore, gangrene does not occur rapidly.

ETIOLOGY**1. Vaso-obliterative Diseases**

e.g. atherosclerosis (senile or pre-senile), vasculitis as Burger’s and Behçet's disease.

- Large Vessel
 1. Takayasu’s arteritis (granulomatous=pulseless dis.)
 2. Giant cell arteritis (temporal arteritis)
- Medium Vessel
 1. Polyarteritis Nodosa (Polyarteritis Nodosa = Kussmaul-Maier disease)
 2. Kawasaki’s arteritis.
- Small Vessel
 1. Wegener’s granuloma,
 2. Microscopic polyangitis,
 3. Henöch-Scholein purpura,
 4. Cryoglobulemic hypersensitivity

2. Specific Forms

- A. **Popliteal artery compartment syndrome** due to compression by the medial head of the gastrocnemius muscle on the artery. Treatment is by myotomy.
- B. **Cystic myxomatous degeneration (Non-atheromatous Popliteal Artery Disease)**
- C. **Carotid stenosis** → atherosclerosis of internal carotid artery→ transient ischaemic attack. Treatment: Thrombendarterectomy and the arteriotomy is closed usually by venous patch. The operation is done using a shunt intra-operative to bypass the site of operation in order not to deprive the brain from its blood supply.
- D. **Subclavian artery stenosis:** Claudication of upper limb. Frank ischemia is rare due to rich collaterals and small muscle bulk of upper limb. It may be associated with cervical rib causing post-stenotic dilatation with artery-to-artery embolization that may cause digit loss

- E. **Subclavian steal syndrome:** The 1st part of subclavian artery is stenosed or occluded. So, shift of blood from carotid artery via circle of Willis down vertebral artery to upper limb during limb exercise resulting in syncopal attack and visual disturbances. Treatment is either PTA, TEA, bypass from the common carotid artery (CCA) to the 3rd part of the subclavian artery.
- F. **Enteric artery occlusive disease “intestinal claudication”** pain after eating, occurs only if 2 of 3 mesenteric arteries are occluded. Treatment is PTA or bypass.
- G. **Renal artery stenosis** cause hypertension in young age not responding to medical therapy. Treatment → PTA, aorto renal bypass, renal artery revascularization by splenic artery.

Predisposing Factors of Atherosclerosis

- Hypertension
- Diabetes mellitus (DM)
- Hyper-lipidemia (hyper-cholesterolemia, hyper-triglyceridemia)
- Obesity
- Stressful lifestyle
- Smoking
- Positive family

CLINICAL PICTURE

The effects of ischemia depend on:

- Type of artery and collateral circulation e.g. subclavian artery *versus* popliteal artery.
- The rate of occlusion of the artery: Acute *versus* chronic ischemia.
- The state of the collateral vessels: Healthy *versus* pathological collateral
- The general condition of the patient: Myocardial insufficiency - severe anemia.

Onset

- The onset of disease is *insidious*, giving time for *collaterals* to develop.

Symptoms

Intermittent Claudication

- Cramp like – by walking – not on 1st step – relived by rest.
- *Site:* The site indicates a step higher occlusion of the artery.
- *Claudication distance* (of prognostic value).
 1. Mild ischemia > 300 m
 2. Moderate ischemia 100-300 m
 3. Severe ischemia < 100 m

Rest Pain

- At foot during rest.
- Increases by lying down or elevation of foot (pressure points) and by foot warming due to increase metabolism → increased O₂ consumption.
- Decrease by foot dependency.
- Differential diagnosis: idiopathic night cramps and peripheral neuropathy in DM

Numbness and Parasthesia

- They result from nerve ischemia.

Trophic Changes

- Painful superficial ulceration between toes.
- Indolent non-healing ulcers in the foot, shin of tibia or around the malleoli.
- Blackened, mummified gangrene.

Impotence

- Impotence may occur due to bilateral internal iliac occlusion as in **Leriche syndrome**

Pre-gangrene

- It is characterized by a combination of rest pain, colour changes, and hyperaesthesia with or without ischemic ulceration.

Signs (Physical Examination)

Inspection

Color

- Limb elevation → blanching (the angle of elevation at which blanching begins is a guide for degree of ischemia called Burger's angle)
- Dependency → purple discoloration due to accumulation of metabolites leading to vasodilatation.

Trophic changes

- Skin is waxy, shiny due to loss of skin appendices.
- Loss of hair and atrophy of nails.
- Ulceration, atrophy and wasting of muscles of affected side.

Palpation

- *Temperature*: Cold, but may be *warm* if infection is superadded.
- *Capillary filling*: Delayed.
- *Venous refilling*: Delayed after emptying the vein by leg elevation for 30 seconds. Guttering of the veins could be seen in severe ischemia.
- *Neurological examination*
 1. Motor: motor loss occurs in acute ischemia, *not* in chronic ischemia.
 2. Sensory: hyperesthesia and pain.

- *Arterial pulsations*

1. Site: Pulsation is absent at site of occlusion downward. Distally the pulsation may be felt due to rich collaterals and can be differentiated by exercise resulting in *disappearing pulse*.
2. Above the site of occlusion pulsation may be stronger due to distal resistance *hammering pulse*.

Auscultation

- Arterial bruit (systolic in nature due to turbulence indicating a proximal stenosis).

Clinical Presentation with Complications

1. *Acute ischemia*

- *Ulceration* of atheromatous plaques may detach and cause distal “*thrombo-embolism*”.
- *Acute thrombosis* on top of the atheroma leads to complete arterial block.
- *Hemorrhage* may occur in the plaque under the intima lead to acute ischemia.

2. *Aneurysm formation* due to weakness of the vessel wall.

Differential Diagnosis of Acute versus Chronic Ischemia (Table 1)

Table 1: Differences between acute ischemia and chronic ischemia

Parameter	Acute Ischemia	Chronic Ischemia
- Age	Young	Old
- History	Cardiac problems	Other manifestations of atherosclerosis
- Source of emboli	Present	Absent
- Preceding claudication	Absent	Present
- Motor function	Absent (no collaterals)	Present

STAGES OF ISCHEMIA

Staging of ischemia according to Fontaine classification and Rutherford classification are shown in Tables 2 and 3, respectively.

Table 2: Fontaine classification

Fontaine	
Stage	Clinical
I	Asymptomatic
II	Intermittent claudication
III	Rest pain.
IV	Trophic changes

Table 3: Rutherford classification

Rutherford	
Category	Clinical
1	Mild claudication
2	Moderate claudication
3	Severe claudication
4	Ischemic rest pain
5	Minor tissue loss
6	Major tissue loss

1. DIAGNOSIS

History-Taking

- By interpretation of clinical finding, the site of occlusion can be predict:
 1. Aorto-bi-iliac occlusion (**Leriche** syndrome): claudication in both gluteal muscles usually begin at one limb at first + impotence and absent pulsation of both common femoral arteries.
 2. Iliac and common femoral occlusions: thigh claudication.
 3. Superficial femoro-popliteal obstruction: claudication in calf.
 4. Tibial vessels: foot claudication.
- In proximal occlusion usually the distal blood supply is good due to presence of rich collaterals, but if symptoms are severe, you have to suspect another distal obstruction (double block).

Investigations

General Investigations

Since atherosclerosis is a systemic disease, the following investigations should be performed:

- Laboratory for DM, hyper-lipidemia, ESR.
- ECG and stress test.
- ECHO cardiography.
- Plain X-ray: may show arterial calcification and flecks of calcium in the wall of atherosclerotic vessels.

Visualization of the Vessels

A. Non-invasive Methods

- *Doppler US* to detect flow and can be used to measure *ankle: brachial pressure index* i.e. systemic pressure in ankle: systemic pressure in brachial artery measured by Doppler and sphygmomanometer.

Resting	0.9-1
Mild ischemia	<0.9
Critical ischemia	<0.3
False high results in case of arterial wall calcification as in DM.....	>1.3

- *Segmental pressure*: can help to locate the site of stenosis.
- *Duplex imaging*: give an idea about vessels blood flow, turbulence & help to know the direction of bl. flow (by coloured duplex).
- *Treadmill with slight incline* (12°) helps to assess walking distance till beginning of claudication.

B. Invasive Methods

- *Arteriography*: done before operation. It shows the arterial tree, the *site* of obstruction and the presence of *run-off* (Table 4).

Types	Technique	Disadvantages
Direct arteriogram	Direct puncture of aorta or carotid	Not done now
Retrograde percutaneous catheterization (<i>Seldinger's technique</i>).	Via femoral or brachial ar.	- Allergy to the dye. - Hematoma. - Infection - Dissection → more occlusion. - Distal embolization due to breakdown of atheroma - Iatrogenic AVF (arterio-venous fistula).
Digital subtraction	Where bone is excluded from the file that offer better visualization of the artery without bone shadow.	
Selective angiography either renal, celiac or carotid.		
Intensified image		
MRA	Clarity is less, but has the advantage of avoiding the need to intra-arterial dye injection or direct catheterization of the artery.	
Recently, endo-luminal U/S (IVUS)		
Multi-slice CTA with virtual 3D construction of the arterial tree		

2. TREATMENT

Intermittent claudication (IC) by itself is *not* an indication for surgery, but interference with normal daily activity, rest pain, ulcer or gangrene are indications for surgery.

A- Conservative Measures

Habits

- Diet (weight reduction and decreasing intake of lipids).
- Stopping smoking completely whether active or passive.
- Foot care as clean foot and protect it from injury.
- Elevation of the head of bed (*Burger's position*) to increase blood flow.

Treatment of Associated Disease

- Diabetes mellitus (DM)
- Hypertension
- Hyper-lipidemia

Exercise

- Walking and **Burger's exercise** by elevation of leg for 2 minutes and then dependency of leg for another 2 minutes to develop collaterals.
- Elevate sole of shoe will ↑ claudication distance as it decreases work on calf muscles.

Drugs

- Aspirin
- Vasodilators: Not of value in atherosclerosis
- Praxilene: Alters tissue metabolism
- Trental (pentoxifylline) helps easy passage of RBCs in capillaries
- Prostacycline: prostaglandin helps vasodilatation, used in critical ischemia, but expensive.

Surgical Treatment

Bypass Surgery (Anatomical)

- Done when there is a good run-off. By using either *saphenous vein* of the patient or a *synthetic material* to bypass the site of occlusion.
- Types of operations
 1. Aorto-femoral bypass (by synthetic graft).
 2. Femoro-popliteal bypass (supra-genicular or infra-genicular) by either *reversed saphenous* bypass *in-situ* saphenous or by *armed PTFE graft* or *in-situ* bypass using a *Miller cuff technique* (constructing the valvotome or synthetic graft anastomosis between the graft and the artery by using an intervening cuff of a vein).
 3. Bypass on *crural vessels* i.e. femoro-post tibial by *in-situ* saphenous graft but has a high rate of thrombosis.

Profundoplasty

- The stenosed profunda is opened and widened by a *vein patch*.

Thrombendarterectomy

- When there is a localized area of narrowing <10 cm in the *proximal vessels*, the atheroma is dissected from the vessel by a dissector and an arterial stripper (loop) is passed between the inner and outer halves of the media.
- *Advantages*: Easy, rapid and does not need graft.
- *Disadvantages*: Leave a rough surface of the artery that may cause thrombosis.

Extra-anatomical Bypass

- Extra-anatomical bypass procedures are used for *limb salvage*
 1. Femoro-femoral cross over graft.
 2. Axillo femoral graft.

Sympathectomy

- It does not improve claudication, but improves *skin circulation* by inducing vasodilatation, and reduces pain.
- *Indications*
 1. Non reconstuctable ischemia (no run-off) and in critical limb ischemia (ulcer or rest pain)
 2. Other indications: (1) Raynaud's disease, (2) Burger' disease, (3) to alleviate pain (cancer bladder, causalgia), and (4) hyperhydrosis (cervical sympathectomy).

Endo-vascular Treatment

- Major operation and surgical morbidity are avoided.
- It gives good results if in *large vessels* in a *confined area*.
- When the patient carries a high risk for operation of aorto-iliac or femoral bypass.
- Methods:
 1. PTA (*balloon dilatation*).
 2. To avoid restenosis, a *stent* can be inserted in large vessels e.g. iliac vessel.
 3. Atherectomy by cutting catheters or laser can be used or rotablattor.
 4. Subintimal tunnling procedure.

Recent Therapy

- New methods of treatment have introduced recently, and include the following:
 1. Gene therapy to modify risk factors.
 2. Stem cell therapy to help neogenesis.

Amputation

- Amputation is resorted to in the following conditions:
 1. Frank gangrene.
 2. Uncorrectable severe ischemia with rest pain and failed other solutions.

ARTERITIS

Thrombangitis Obliterans (Burger's Disease)

Characteristics

- It is an obliterative disease, characterized by:
 1. Occlusive disease of small and medium-sized *arteries*
 2. Thrombo-phlebitis of superior and deep veins
 3. Raynaud's phenomenon ($\pm$)
 4. Pan-arteritis i.e. affecting all layers $\rightarrow$ thrombosis

Etiology

- *Unknown*, but it is thought to be an *autoimmune* process aggravated by *smoking*

Clinical Picture

- The disease has an episodic course, each is stronger than the previous one.
- It occurs more in males, middle-aged, and smoker.
- Signs and symptoms of chronic arterial occlusion
- Affects lower limbs much more than upper limbs.
- The disease may be preceded by migrating thrombophlebitis, which appears as a small, tender cord along the course of a vein.
- Pedal pulses are usually absent.

Pathology

- It affects the popliteal artery and the vessels distal to it.
- The neuro-vascular bundle shows an inflammatory reaction (pan-vasculitis and neuritis)

Investigations

- *Arteriogram* shows affection of small- and medium-sized artery with well-developed collaterals (*cork screw*), no calcification or irregular plaques (differential diagnosis: atherosclerosis).
- *Biopsy* is the surest method of diagnosis. It shows infiltration of the arterial wall, vein and nerves by inflammatory cells occluding the lumen, but it is *not* done.

Differential Diagnosis (Table 5)**Table 5:** Differences between atherosclerosis and Burger's disease

Parameter	Atherosclerosis	Burger's Disease
<i>Etiology</i>	A degenerative disease.	Immunological process, potentiated by smoking.
<i>Pathology</i>		
- Artery	Proximal	Distal
- Vein	No	Yes
- Nerve	No	Peripheral nerve involvement
<i>Clinically</i>		
- Sex	♂ > ♀	About to occur only in ♂
- Age	Old (60 years + DM)	40 years
- Smoking	± smoker	Heavy smoker
- Gangrene	Can be extensive.	Usually limited to toes.
<i>Investigations:</i> <i>Arteriogram</i>	Calcification. - Big-sized vessels. - Plaques (irregular lining). - Gradual lesion. - Collaterals.	No calcification. - Small size vessels. - Smooth intimal lining. - Abrupt lesion. - Collaterals (<i>cork-screw</i>)

Treatment

- Stopping smoking (the most important).
- Anti-platelets.
- Steroids.
- Vasodilators, but some authors deny its effect as it may cause ‘steal phenomena’.
- Lumbar sympathectomy.
- Conservative amputation, if needed.

Other Types of Arteritis

Takayasu’s Arteriopathy

- Granulomatous = pulseless disease
- More in women
- Affects large vessels

Giant Cell Arteritis (temporal, occipital and ophthalmic arteritis)

- It affects large- and medium-sized blood vessels (cranial branches of aorta).
- It is most often seen in patients over 50 years of age, more in females
- *Clinical picture*
 1. Headache localized to the area overlying the superficial temporal arteries.
 2. Sudden blindness.
 3. Scalp tenderness
 4. Jaw claudication
 5. Tongue claudication and dysphagia.
 6. Tender palpable occluded vessels
- *Diagnosis* is by biopsy, which shows ‘granulomatous arteritis’
- *Treatment* by corticosteroids

Polyarteritis Nodosa (Polyarteritis Nodosa = Kussmaul-Maier Disease)

- It is a systemic *necrotizing* inflammation of blood vessels (vasculitis) affecting *medium-sized muscular* arteries, typically involving the arteries of the *kidneys* and other internal organs.
- *Gender*: more common in men,
- *Age*: ranges between 40-60 years.
- *Clinical picture*
 1. Renal disease occurs with progressive renal failure.
 2. Hypertension
 3. GIT involvement as abdominal pain, nausea and vomiting.
- *Investigations*: elevated ESR, angiography may be diagnostic showing the characteristic findings of saccular or fusiform aneurysms and arterial narrowing.
- *Treatment*: corticosteroids and other immuno-suppressants.

Behçet's Disease

- *Gender*: male to female ratio is 4:1
- It is characterized by its potential to affect *veins*, as well as *large, medium, and small* arteries causing:
 1. Deep vein thrombosis (DVT)
 2. Aneurysm
 3. Acute or chronic arterial occlusion.

NON-ATHEROMATOUS POPLITEAL ARTERY DISEASE

Cystic Myxomatous Degeneration

- Accumulation of clear jelly-like material in the outer layers of the main artery, especially in the popliteal artery.
- Incidence: mid-thirties, more in Asia (Japan).
- Sudden onset of short-distance calf claudication.
- Claudication symptoms may completely resolve for a period of time and then recur, or they may progress rapidly
- Pulsations disappear on flexing the knee (Ishikawa's sign),
- Audible bruit in the popliteal fossa.
- Arteriogram shows smooth narrowing of a normal artery with narrowing or occlusion on flexion.
- Treatment: removal of myxomatous material, if recurred needs resection of the artery.

Popliteal Artery Compartment Syndrome

- *Etiology*: Compression by the medial head of gastrocnemius muscle on the artery
- *Treatment*: Myotomy.

VASOSPASTIC CONDITIONS

Etiology

Primary Raynaud's Phenomenon (Disease)

- There is no known cause to abnormal sensitivity of arterioles to cold. Other stimuli than cold can precipitate attacks like: tobacco & smoke. In addition to the digits, the vasospasm can involve the nose, ear lobes and exposed skin.
- It affects young women, affecting upper limbs more than lower limbs.
- Exposure to cold → constriction of arterioles → *Blanched* → capillaries dilate and become filled with slowly flowing deoxygenated blood → Swollen and dusky colour → arterioles relax → oxygenated blood → Red painful hands ± superficial necrosis of tips of fingers.

Secondary Raynaud's Phenomenon

- Vasospasm occurs secondary to the following:
 1. Connective tissue disease: systemic lupus erythematosus (SLE), scleroderma, etc.
 2. Carpal tunnel syndrome and cervical rib.
 3. Workers working with tools that vibrate at certain frequencies e.g. pneumatic drills.
 4. Atrophic changes in the limb (poliomyelitis, following trauma, etc).
 5. Frost-bite.

Investigations

- *Duplex* → normal waves till radial and ulnar arteries.
- *Measuring the digital systolic pressure* as base line and after cooling to 15°C. It is diagnostic if drop in pressure is > 30.
- *Cryogenic arteriography* shows spasm after exposure to cold.
- *Investigations to exclude 2^{ry} causes*: ↑ ESR, antinuclear Ab, anti-double strand DNA Ab, rheumatoid factor, Anti CCP, cryo precipitate, cold agglutinin and chest X-ray.

Differential Diagnosis

- *Primary versus secondary types* (Table 6)

Table 6: Differences between primary and secondary Raynaud's phenomenon

Parameter	Primary	Secondary
- Gender	More in females	Equal incidence
- Side	Bilateral	Unilateral
- Another disease	Absent	Present

- *Ergotism*: used for treatment of migraine, post-partum hemorrhage, eating rye bread infected with fungus.
- *Cryoglobulinemia*: It is a medical condition in which the blood contains large amounts of pathological cold sensitive antibodies called cryoglobulins – proteins (mostly immunoglobulins themselves) that become insoluble at reduced temperatures

Treatment

- Avoidance of cold.
- Calcium-channel blockers such as nifedipine.
- Cervical sympathectomy if all fails.

ENDOVASCULAR SURGERY

Definition

Endovascular surgery is the management of vascular diseases percutaneously through a puncture in the vessel (artery or vein) to deal with a lesion in a remote site via the lumen of the vessel.

Purpose

1. Diagnosis.
2. Treatment
3. Prophylaxis

Advantages

- Minimally-invasive, mostly performed under local anesthesia
- Short hospital stay, quick recovery time and early return to work
- Can reach difficultly accessed areas.
- Avoids incision in areas of infection.

Disadvantages

- Hazard of radiation (for patient and medical staff).
- Dye-induced nephropathy.
- May be higher in cost.
- Patency rate is < surgery in certain situations.
- Stent fracture or stenosis.
- Vessel dissection or perforation.
- Access site complications such as hemorrhage, pseudo-aneurysm, hematoma or arterio-venous fistula (AVF) formation.

Requirements

- Contrast media
- Needle, catheter sheath introducer.
- Guide wire, selective catheter, flush catheter, guiding catheter
- Balloon, stent, inflator.

Type of Access

- The access could be *retrograde*, *antegrade* or *contralateral* (to avoid hostile and difficult groin in obese patients).

Balloon Angioplasty (Percutaneous trans-luminal angioplasty = PTA)

Indications

- Indications are the same as for conventional vascular operations i.e. critical ischemia and severe incapacitating claudication.

Mechanism of Action

- Cracking of the atheroma; thus allowing vessel dilatation and increased blood flow.
- Squeezing some of the core of the atheroma; thus decreasing its volume.

Types

- *Plain balloon* (bare balloon).
- *Drug-eluting balloon*: It has cytotoxic drugs that help to decrease myoepithelial hyperplasia and hence decrease re-stenosis
- *Cryoplasty balloon*: It also aims at decreasing myoepithelial hyperplasia and hence decrease re-stenosis.
- *Cutting balloon*: It has sharp edges to cut resilient fibrotic lesions.

Endoluminal Stents

- Endoluminal stents help to maintain the vessel patent and are used to treat dissection, restenosis or residual stenosis.
- Types of stents
 1. *Balloon-mounted stent* (high radial force, precise deployment, but rigid i.e. difficult to deployed via a severely tortuous artery).
 2. *Self-expandable stent* (less radial force, less precise deployment, but more flexible i.e. good for a severely tortuous artery).
 3. *Drug-eluting stent* (DES) has cytotoxic drugs that help to decrease myoepithelia
 4. al hyperplasia and hence decrease re-stenosis.
 5. *Biodegradable stent*. It degrades with time to prevent metallic burden on the vessel wall and so decreases re-stenosis.

Debulking Techniques

- Debulking techniques aim at removal or decreasing atheroma of the vessel.
- These techniques are expensive and may cause perforation.

Intra-vascular Ultrasound (IVUS)

- Introducing intra-vascular probe of US to detect detailed internal criteria of the vessel and atheroma.

Indications of Endovascular Treatment

Diagnosis of Vascular Diseases

Treatment

1. Arterial Diseases

- a) *Arterial dilatation (aneurysm and pseudo-aneurysm)*:
 - Small artery (occlusion via coils or thrombin injection).
 - Big artery (covered stent, endovascular aortic repair EVAR).
- b) *Chronic ischemia*
 - Treatment of occluded vessel (peripheral, carotid, renal or mesenteric) by PTA, stent or debulking devices.

- c) *Acute ischemia*
 - Thrombo-aspiration or thrombolytic therapy via a catheter applied intra-lesionally
- d) *Selective occlusion of an artery*
 - Fibroid post-partum hemorrhage.
 - Treatment of malignant tumor by feeding arterial artery occlusion or bullet chemotherapy selectively in an artery feeding the tumor.

2. Venous Diseases

- a) Traumatic AVF with occlusion of the rent by covered stent.
- b) SVC or IVC occlusion stenting.
- c) Venous stenting to treat vein compression by iliac artery (*May Turner syndrome*).
- d) Deep vein thrombosis (DVT) treatment thrombolytic therapy.
- e) Retrieval of lost wire or catheter during intervention (iatrogenic) by snare.
- f) Varicose vein (VV) treatment (radiofrequency) or LASER ablation).
- g) Insertion of inferior vena cava (IVC) filter.

3. Cardiac Diseases

- a) Radiofrequency ablation of ectopic foci for treatment of arrhythmia and pace maker.
- b) Annular valve dilatation, valve repair by clipping or trans-catheter aortic valve replacement (TAVR) or trans-catheter aortic valve implantation (TAVI).
- c) Ventricular septal defect (VSD) closure.

Inferior Vena Cava Filter

- The filter does not prevent pulmonary embolism (PE), but prevents major emboli that could be life-threatening.
- Indications
 1. DVT or documented PE in a patient who has a *contra-indication* to anti-coagulations, e.g. previous intra-cerebral hemorrhage or history of bleeding peptic ulcer.
 2. Recurrent PE despite adequate anti-coagulation.
 3. Complications of anti-coagulation that forced therapy to be discontinued.
 4. Poly-traumatic patients where liability of the use of anti-coagulants is *not* feasible while the patient is at high risk of developing DVT due to hyper-coagulable state as a response to trauma “temporary or retrievable filter is used”.
- Complications
 1. Improper placement, tilting, angulation and migration
 2. Filter deformation, fracture or penetration
 3. Superadded thrombosis.

CHAPTER

3

Arterio-Venous Fistula (AVF)

Definition

It is a form of *acquired* abnormal connection between the arterial and venous sides of the circulation. It has the effect of short-circuiting the normal anatomic pathway of blood flow away from the peripheral vascular bed.

ETIOLOGY

1. Traumatic

- It is the most common cause of AVF.
- Usually due to penetrating injury.

2. Therapeutic

- Hemodialysis.

3. Pathological

- Neoplasms (neoplastic invasion).
- Paget's disease of the bone.
- Inflammatory e.g. an abscess eroding nearby vessels - Aneurysm eroding nearby vessels.

4. Iatrogenic

- Operative
- Catheter-induced

TRAUMATIC ARTERIO-VEINOUS FISTULA

Etiology

1. It is usually the result of a *penetrating* injury (stab 63%, bullet or gunshot 26%) or rarely, blunt trauma (1%). The commonest vessel is the femoral artery.
2. Carotido-cavernous fistula may be the result of *fracture base* of the skull
3. Iatrogenic:
 - AVF between iliac artery and vein during removal of an inter-vertebral disc or gynecological procedures.

- During thyroidectomy, nephrectomy, splenectomy or amputation. If the artery and vein are ligated together, the ligature might erode through both vessels and communication occurs (very rare).
- After percutaneous catheter maneuvers

Patho-physiology (Figure 1)

Patho-physiological changes that occur in case of AVF are shown in Figure 1.

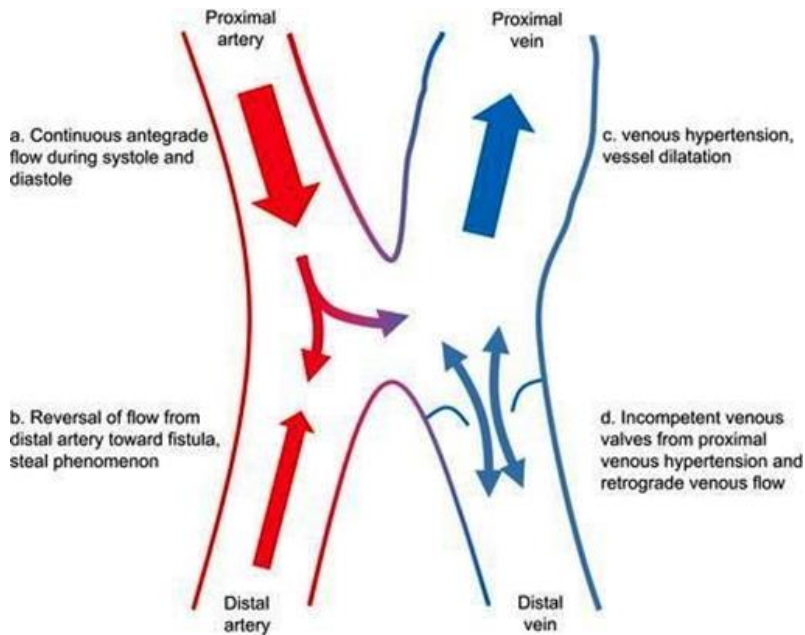


Figure 1: Patho-physiology of arterio-venous fistula

Changes depend on:

1. Site of the fistula “proximal or distal”.
2. Magnitude of blood flow.
3. Resistance of the fistulous bed.
4. Diameter of connecting channel.
5. Length of the channel.

Changes will affect the following:

- *Proximal (afferent) artery:* flow in the artery is normal with thinning of the wall due to atrophy of medial smooth muscle.
- *Distal (efferent) artery:* ↑ size of the fistula → ↓ resistance through the fistula and reversal of flow to the site of fistula with distal ischemia.
- *Proximal vein (efferent):* ↑ flow in it in the direction to the heart, with *arterialization* of vein with dilatation and tortuosity.
- *Distal vein:* ↑ flow away from heart that results in dilatation and incompetent valves → varicose veins and venous hypertension with progressive distal venous insufficiency.
- *Fistulous channel:* Dilates with time and may → local aneurysm in the channels or vein.
- *Collateral vessels:* Develop very rapidly on both, venous and arterial sides.
- *Regional effects:* pulsatile swelling and dilated venous channels.

Clinical Picture

- *Duration*: Traumatic AVF usually develops either in one week (two-thirds of patients) or takes long time up to years to develop (one-third of patients).
- *History-Taking*: Trauma (whether recent or old), surgery, intra-arterial catheterization.
- *Vital signs*
 1. Blood pressure (BP): $\uparrow$ systole (due to $\uparrow$ venous return) + $\downarrow$ diastole (due to $\downarrow$ peripheral resistance) $\rightarrow$ $\uparrow$ pulse pressure.
 2. Heart rate (HR): Tachycardia.
- *General examination (Systemic findings)*
 1. Heart: (1) $\uparrow$ cardiac inflow and venous return, (2) cardiac hypertrophy with a systolic murmur (3) Congestive heart failure (CHF) only in large long-standing fistulae.
 2. Pulmonary edema.
 3. *Branham's sign*: Slowing of the pulse rate when the fistula is obliterated by the finger as shown by disappearance of the murmur (relative bradycardia)
- *Local signs*
 1. Pulsating swelling (Figure 2).
 2. Warm skin in the region involved ($\uparrow$ local temperature).
 3. Palpable thrill and audible continuous machinery murmur systolic and diastolic (*characteristic*).
 4. Prominent dilated veins (Figure 3).



Figure 2. AVF presenting with a pulsating swelling



Figure 3. AVF presenting with dilated tortuous veins over the right upper limb and chest wall

- *Distal signs*
 1. Dilated, thick-walled pulsating veins. Pressure on the site of the fistula $\rightarrow$ collapse of those dilated veins completely.
 2. Venous hypertension $\rightarrow$ varicose veins (VVs) with peripheral pigmentation, edema, ulceration, resulting from venous insufficiency.
 3. Signs of arterial insufficiency in the limb, weak or absent pulses (*steal syndrome*).

Investigations (Diagnostic Aids)

1. *Duplex Scan*
 - Loss of the typical tri-phasic arterial flow pattern
 - Arterialization of venous flow signals
 - Reversal of flow in the distal artery
 - Turbulence flow at site of fistula
 - Mosaic appearance
2. *CT-angiography*
3. *Angiography* (Figure 4)
 - Localize the fistula site Angiography:
 - Visualization of both arteries and veins
 - It can detect associated lesions such as aneurysms
4. *MRA (MR angiography)*



Figure 4: Angiography showing a traumatic arterio-venous fistula (AVF)

Differential Diagnosis

AVF should be differentiated from congenital arterio-venous malformation (AVM) (Table 1) and from aneurysm (Table 2)

Table 1: Differences between congenital arterio-venous malformation and traumatic AVF

Criteria	Congenital AVM	Traumatic AVF
<i>History of trauma</i>	-ve	+ve
<i>Locally</i> - Area affected - Continuous murmur and dilated veins - Gigantism of the limb <i>Branham's sign.</i>	Diffuse areas -ve ± -ve	Localized +ve Fades completely on digital pressure -ve +ve
<i>Treatment</i>	Palliative, difficult, indefinite	Correctable

Table 2: Differences between AVF and aneurysm

Criteria	AVF	Aneurysm
<i>Etiology</i>	Commonest is traumatic	Commonest is atherosclerotic
<i>Locally</i>	Swelling is less marked Murmur: systole-diastolic (Continuous machinery murmur)	Well marked Systolic only
<i>Distally (Varicosities)</i>	+ ve	-ve
<i>Central</i>	Tachycardia/Branham's Heart failure	-ve -ve

Treatment

- Iatrogenic (if small): compression-guided ultrasound
- Traumatic (non-iatrogenic or gunshot): early repair

Options of Treatment

- Endovascular
- Open surgical repair
- Hybrid technique
- Combined endo- and open surgery especially in inaccessible sites or large fistula

Indication of Intervention

- Hemorrhage
- Venous or arterial insufficiency
- Cosmetic deformity.
- Heart failure.

Endovascular Procedures

- Embolization using coils or gel foam
- Endo-graft (stent graft)

Open Surgical Repair*Fistulous connection interruption*

- Ligation
- Repair of the involved artery and vein

Arterial reconstruction

- Patch
- Interposition graft

If infected

- Ligation only
- Ligation with extra-anatomical bypass

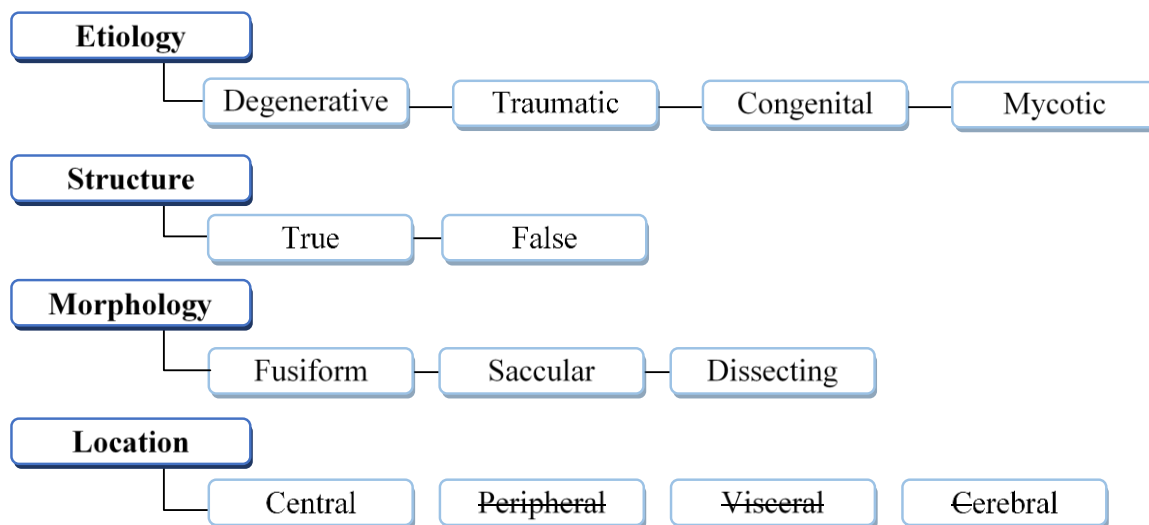
Arterial Aneurysms

Definition

Aneurysm means "*permanent localized dilatation* of blood vessel >1.5 times of its original diameter". It is derived from the *Greek* word "aneuryn", which means "*dilatation*". It may affect arteries (more common) or veins. If *generalized*, it is known as *ectasia*

CLASSIFICATION

Aneurysms can be classified according to *etiology* (degenerative, traumatic, congenital and mycotic), *structure* (true or false), *morphology* (fusiform, saccular or dissecting), and *location* (central, peripheral, visceral, cerebral) as shown below.



According to Etiology

1. Degenerative

- It is the commonest type of aneurysms and the commonest cause is *atherosclerosis* leading to aneurysmal formation at various sites e.g. abdominal aorta and popliteal artery.
- Predisposing factors include old age, obesity, hypertension, smoking and male gender

- Aneurysms are caused by weakness of the arterial wall due to continuous break down of elastic and collagen fibers coupled with decrease in vasa vasorum.

2. Traumatic

- A *blunt trauma* to an artery may weaken part of its wall. Later, this weak area progressively yields leading to aneurysmal dilatation. Acceleration and deceleration in cars → thoracic aortic aneurysm, and palmer artery aneurysm in manual workers.
- *Penetrating injury* to an artery by needles, catheters, or stabs may cause a small hole (puncture) in the wall & → hematoma surrounding the artery. Later on, this clot is surrounded by a false capsule of organized fibrous tissue (FT) & the result will be a *false aneurysm* (usually saccular in shape).

3. Mechanical

- *Post-stenotic dilatation*: Turbulence of blood → aneurysm formation with further ↑ of turbulence → more ↑ in aneurysm size (vicious circle) e.g. cervical rib.

4. Congenital aneurysm

- These may occur in the circle of **Willis** where they may cause *sub-arachnoid* hemorrhage e.g. **Berry** aneurysm in cerebral vessels.
- Other sites include the splenic, renal or celiac vessels.

5. Collagenic

- Behcet's disease
- Marfan's syndrome and Ehlers Danlos syndrome.

6. Inflammation and infection

- *Mycotic* (A misnomer as it caused by *bacterial* infection, *not* fungal) e.g. bacterial endocarditis or salmonella infection.
- Syphilis: end arteritis of the vasa vasora.
- Pan (poly) arteritis nodosa (PAN) and peri-arteritis.

According to Structure (Nature)

1. True Aneurysm

- The wall formed of all 3 layers of the artery

2. False Aneurysm

- The wall formed by hematoma, fibrous tissue (FT) or part of wall e.g. following trauma (hematoma that communicates with an artery via a rent), or in case of dissecting aortic aneurysm (usually due to hypertension).

According to Morphology (Shape)

1. Saccular

- Arises from side of artery (Figure 1)

2. Fusiform

- Diffuse, symmetrical e.g. atherosclerosis and syphilis (Figure 1)

3. Dissecting

- Between two halves of media causes e.g. atherosclerosis, medial necrosis and **Marfan** syndrome.

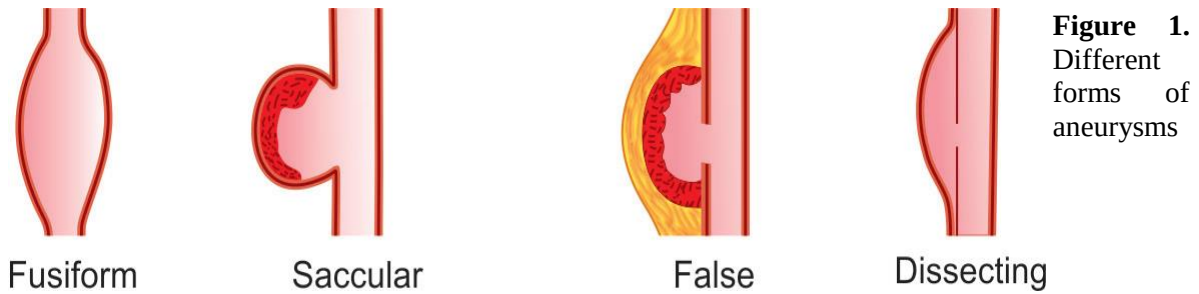


Figure 1.
Different
forms of
aneurysms

According to Location

1. *Central*
2. *Peripheral* e.g. carotid artery or femoral artery aneurysm.
3. *Visceral* e.g. splenic or renal artery aneurysm.
4. *Cerebral* e.g. *Berry* aneurysm

PATHOLOGY AND PATHOPHYSIOLOGY

Pathology

- *True aneurysms*: The wall is formed of the 3 layers of the dilated artery.
- A *false aneurysm* (Figure 2) is actually a hematoma communicating with lumen of an artery through a partial tear in its wall. Thus, the wall of the false aneurysm is formed by the fibrous wall of the hematoma and the surrounding tissue.

Pathophysiology

- Dilatation of the artery changes the flow pattern of the blood from laminar flow into eddies, which → formation of a thrombus, which deposits on the wall causing its thickening.
- The thrombosis may extend distally and may fragment leading to distal embolization and hence distal ischemia.



Figure 2: Pseudo-aneurysm of the brachial artery

COMPLICATIONS

1. Leakage (Figure 3) / Rupture (Figure 4)

- It is the most serious complication.
- It may cause *fatal hemorrhage* (anterior rupture) or retro-peritoneal leakage (hypotension + pressure in that retro-peritoneal space stops bleeding (posterior rupture))

2. Thrombosis / Embolization (Distal Ischemia)

- Thrombosis and occlusion of the aneurysm.
- Detachment of fragments of this thrombus cause distal *embolization* e.g. in the feet (Figure 5) or gangrene in one of the fingers due to distal emboli from subclavian aneurysm causing post-stenotic subclavian artery aneurysm in patients with cervical rib.

3. Infection (Figure 6)

- Infection may lead to rupture and 2^{ry} hemorrhage.

4. Compression on Adjacent Structures

- Compression of an adjacent *vein* may → venous obstruction or thrombosis.
- Compression of a *nerve* may → motor or sensory affection.
- The adjacent *bone* may also be eroded e.g. lumbar vertebrae in huge abdominal aortic aneurysm (AAA) (Figure 7).

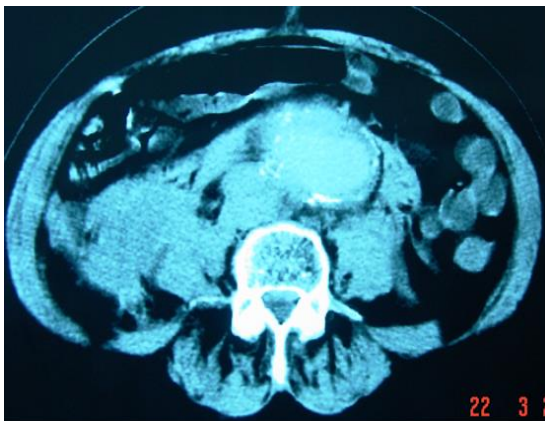


Figure 3. CT showing leakage from aneurysm.



Figure 4. CT showing rupture of aneurysm.



Figure 5. Embolization of aneurysm (gangrene)



Figure 6. Infection of left popliteal aneurysm

5. Complications of Dissecting Aneurysm

- Weakening of the outer wall of the aorta → rupture or formation of an aneurysm.
- Occlusion of aortic branches → myocardial infarction, stroke, renal failure (RF), bowel ischemia, paraplegia or limb ischemia.
- Disruption of the aortic valve annulus → acute valvular insufficiency (acute aortic regurge) and heart failure (HF) or coronary ostea (angina pectoris).
- Rupture either internal or external (pericardium or mediastinum).
- Occlusion of renal vessels → renal failure
- Occlusion of iliac or femoral artery → lower limb ischemia.



Figure 7. Large aneurysm causing compression on the vertebrae.

CLINICAL PICTURE

Characteristic Features

1. *Silent* aneurysms. Some aneurysms, e.g. of the abdominal aorta are commonly silent.
2. *Accidentally* discovered at clinical or US examination done for another reason.
3. Symptoms due to *compression* on adjacent structures.
4. *Presentation by complications*.
5. *Local signs* of an aneurysm

Symptoms are According to Site

Thoracic

1. Asymptomatic.
2. Severe chest pain (intra-scapular) (*tearing character*).
3. Dysphagia.

Abdominal

1. Upper abdominal pain, low back pain.
2. Gastrointestinal (GI) or urinary pain due to compression.
3. Ischemia of LL.

Symptoms of Complications

1. Fainting attack & back pain due to leaking or rupture of the aneurysm, which may cause sudden onset of pain, shock and abdominal distention.
2. Horner's syndrome.
3. Disturbed sensorium (sub-arachenoid hemorrhage).
4. Hoarseness of voice.

Signs (Physical Examination)

Aneurysm (Swelling)

- A *smooth* swelling (Figure 8) that lies along the line of an artery and can be moved across the line of the artery but *not* along it.
- The swelling exhibits *expansible* pulsations (i.e. expansion in *all* directions).
- Proximal pressure on main artery → ↓ or disappearance of pulsations in the swelling.
- A systolic *thrill* may be felt and a *bruit* may be heard over the swelling.
- If you can palpate the *upper limit* of aneurysm below xiphi-sternum with the patient lying flat, it is usually an *infra-renal* aneurysm.

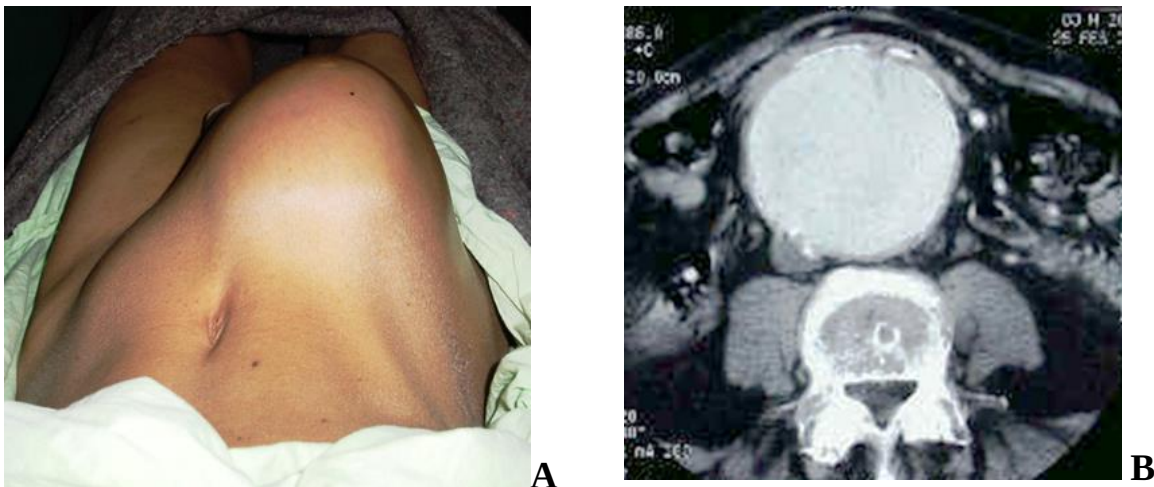


Figure 8. Large pulsatile abdominal mass (A) seen on abdominal CT scan (B)

Signs of Pressure (Compression)

- Sternum → erosion of the bone.
- Nerves → neurological pain.
- Veins → edema and may cause thrombosis.
- Vertebra → erosion of the bone but *not* the disc.
- Trachea → dyspnea.
- RLN → diminished vocal cord motility
- Sympathetic chain → Horner's syndrome.

Signs of Complications

- *Retro-peritoneal leakage* → hypotension.
- *Distal ischemia* → gangrene in toes or fingers.
- *Infection* → signs of acute infection (redness, tenderness, etc).
- *Compression* on adjacent structures e.g. on nerves → motor or sensory affection.

DIFFERENTIAL DIAGNOSIS

- A **swelling** may be:
 1. Over an artery e.g. pseudo-pancreatic cyst: *Transmitted pulsation* (Figure 9).
 2. Below the artery e.g. cervical rib or LN: The mass may be felt.
 3. *Pulsating* e.g. osteogenic sarcoma, osteoclastoma, 2^{ry} to hypernephroma, or thyroid cancer: Other manifestations + ill-defined border + *not* in line with the artery.

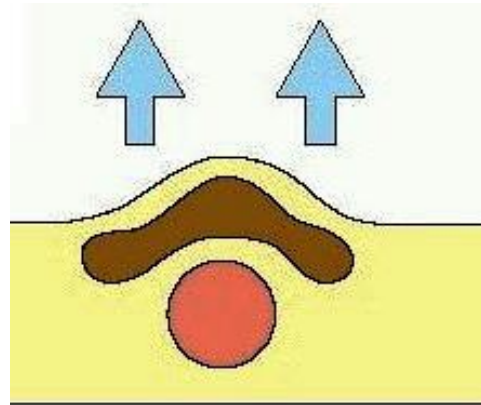


Figure 9. Transmitted pulsations

- An aneurysm (expansile pulsation – Figure 10) should be differentiated from the following
 1. A swelling overlying an artery may elicit transmitted arterial pulsations – not expansile pulsation-. Pressure on the proximal artery does *not* change the size of the swelling. If the swelling can be moved away from the artery, the pulsations disappear.
 2. Abscess (in case of infected aneurysm).
 3. Mass under the artery
 4. A very vascular tumor e.g. osteogenic sarcoma or pulsating metastases.
 5. An abscess is tender, not compressible & not pulsatile.
 6. An *ectatic artery* (carotid knuckle)) - arteriogram
 7. *Cirroid aneurysm of the scalp*: It is a traumatic AVF from birth trauma or unnoticed trauma. It may be congenital. C/P: pulsating swelling in scalp, compressible, ± bruit and may erode underlying bone.
 8. *Arterio-venous fistula (AVF)*: to-and-fro machinery murmur, 2^{ry} varicose veins and positive *Brenham's* sign.

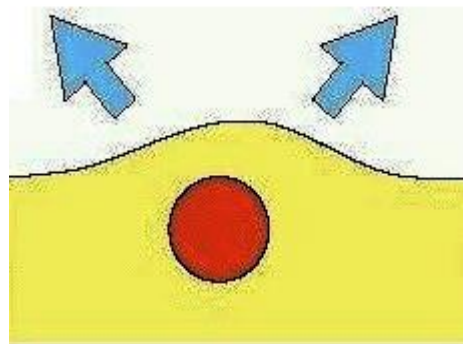


Figure 10. Expansile pulsations

DIFFERENTIAL DIAGNOSIS

1. General laboratory investigations
2. Tests to exclude connective tissue diseases.
3. Plain X-ray shows a soft tissue shadow (Figure 11).
4. Duplex scanning: It shows the aneurysm itself and may also detect a mural thrombus (Figure 12).



Figure 11. Plain X-ray abdomen showing a large soft tissue shadow (red arrows)



Figure 12. Duplex scan of brachial artery aneurysm showing mural thrombus

5. *CT scan with contrast:* It is a very accurate in assessing the diameter of the aneurysm (Figure 13) and in diagnosis of leakage (Figure 14) or rupture of thoracic or abdominal aneurysms.



Figure 13. CT scan can detect and measure the aneurysm (arrows)

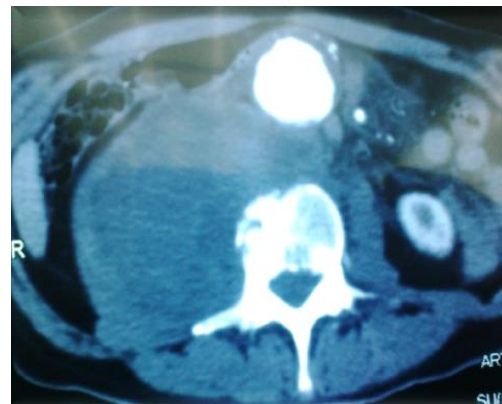


Figure 14. CT of abdominal aorta showing leaking aneurysm

6. *Arteriography / CT angio* (Figure 15): It may be needed to visualize the distal arteries to determine the extent of the aneurysm (e.g. AAA). It is *not* used to measure the size of the aneurysm. The catheter is introduced via the brachial artery and *not* the femoral artery to avoid the possibility of rupture of the aneurysm. Recently, 3D-CT scan has been introduced and provide more details about the aneurysm (Figure 16).



Figure 15. CT angiogram showing abdominal aortic aneurysm (AAA)



Figure 16. 3D-CT angiogram showing abdominal aortic aneurysm (AAA)

7. Magnetic resonance imaging scan (MRI) (non-invasive as opposed to angiography).
8. Digital subtraction angiography (DSA)

TREATMENT

Aneurysms are liable to rupture and should be treated *surgically* (endovascular or open by ligation, excision or bypass).

1. The standard line of treatment is *excision and grafting* (Figure 17A-B).
2. Small aneurysm in small arteries → *ligation* or trial of inducing thrombosis in the aneurysm by fine wire application, via *interventional radiology*.

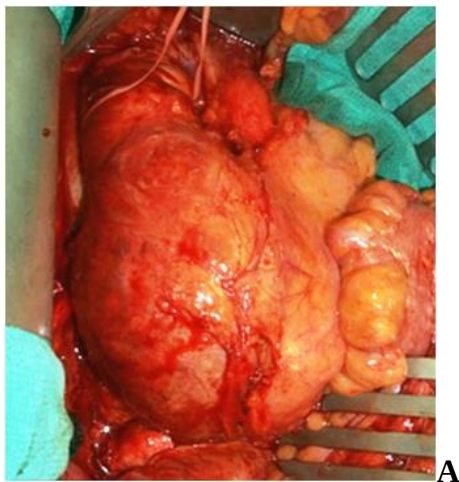


Figure 17. Abdominal aortic aneurysm (AAA) in a patient (A) treated with excision and grafting (B)

3. *Aneurysmorrhaphy*
 - Excising part of the aneurysm and repairing the wall. *Not done now.*
4. *Exclusion graft ((in-lay graft)*
 - Insertion of a graft *inside* the sac of the aneurysm without removal of the sac, this is commonly done in aortic and popliteal aneurysms.
5. *Treatment of abdominal dissecting aneurysm*
 - Conservative treatment by anti-hypertensive drugs, but if increasing in size, causing neurological manifestations, or having an acute presentation → replacement or intimal fixation.

SPECIAL TYPES OF ANEURYSMS

Popliteal Aneurysm

- It may be bilateral
- It may be associated with aortic aneurysm.
- *Diagnosis:* CT arteriography (Figure 18)
- *Complications:* Acute thrombosis & distal embolization.
- *Treatment:* Bypass graft

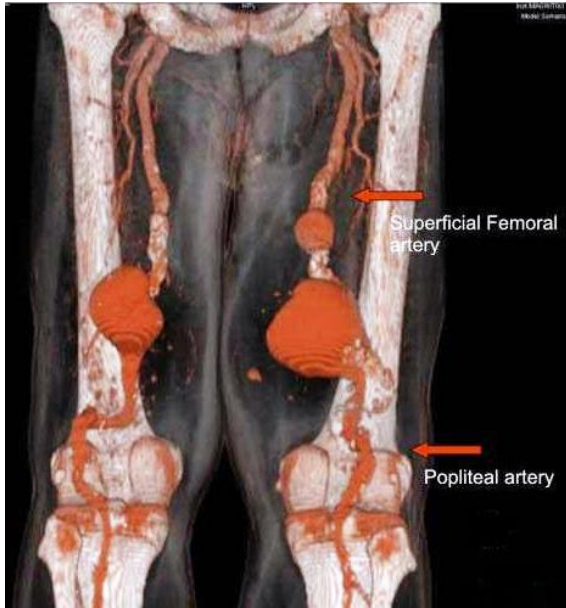


Figure 18. CT arteriogram showing popliteal artery aneurysm

Femoral Aneurysm

- *True aneurysm:* It is uncommon, not liable to complications and treatment is conservative.
- *False aneurysm:* It is mostly infective or traumatic. Currently, the most common cause is *iatrogenic* after PTCA or PTA. It needs reconstruction or may be treated by compression therapy guided by Duplex in case of small neck pseudo-aneurysms.

Sciatic (Persistent) Artery Aneurysm

- Persistent sciatic artery is a rare (0.03-0.06% of the total population) congenital vascular anomaly. During embryogenesis, the sciatic artery runs along the sciatic nerve and comprises the major blood supply to the respective lower limb. As the femoral artery is formed, the bilateral sciatic arteries start to regress. This process continues until the femoral artery completely takes over as the major lower limb blood supply, leaving only remnants of sciatic artery.
- A persistent sciatic artery may occur when there is either a failure of involution of the sciatic artery or failure of the proper normal development of the femoral artery. Most cases of persistent sciatic artery are clinically silent and detected only after a vascular complication. Aneurysms develop in approximately 25% of cases, likely due to the vessel's anatomical predisposition to minor trauma with sitting. Other complications include thrombosis, embolism and rupture of the artery causing ischemia and pain of the lower limb.
- *The preferred diagnostic modalities* are US and CT scan. The CT angiography (Figure 19) is useful to delineate the proximal and distal anatomy of the vessel and assist in pre-operative planning.
- Treatment options include surgical exclusion or excision of the aneurysm, endovascular stenting and endovascular coiling.



Figure 19. 3D-CT arteriogram showing sciatic artery aneurysm

Iliac Aneurysm

- It is rarely found "isolated".
- More commonly, it may be associated with an aortic aneurysm

Abdominal Aortic Aneurysm (AAA)

Incidence: It is the most frequent type of aneurysms.

Site

- *Infra-renal* in 95% of cases (Figure 20). It may extend to affect the *iliacs*.
- Rarely, it extends *upwards* to involve a variable distance of the *supra-renal* abdominal aorta or the thoracic aorta in which case it is called (*thoraco-abdominal aortic aneurysm*).

Etiology: Degenerative disease is the commonest cause.

Clinical Features

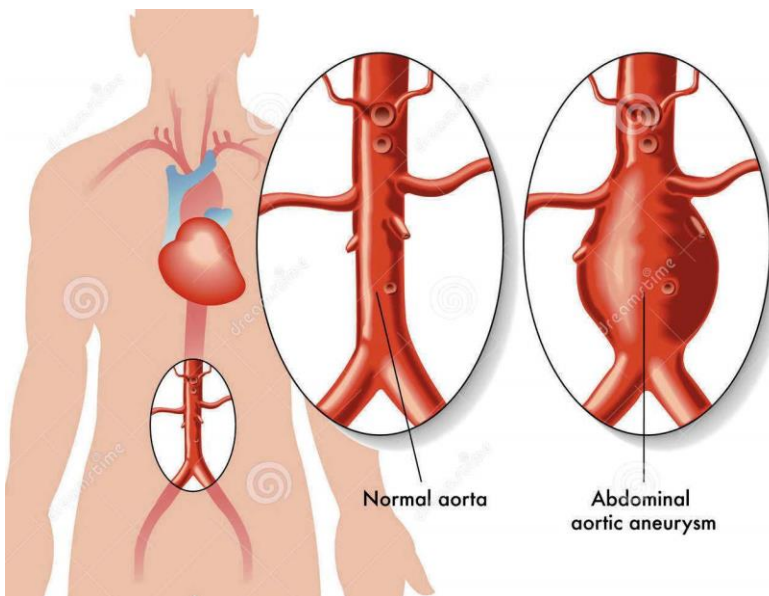


Figure 20. An abdominal aortic aneurysm (AAA) is infra-renal in the majority of cases

1. *Asymptomatic aneurysms.* In 75% of patients the AAA is discovered accidentally during a routine abdominal examination (as a pulsatile epigastric mass), or during a radiographic study performed for some other reason (plain X-ray, US or CT scan).
2. *Pain* is the commonest symptom. An AAA gradually enlarges and impinges on surrounding structures causing vague abdominal pain. Back and flank pain result from vertebral compression. Large aneurysms can erode the spine and cause severe back pain in absence of rupture. These patients may be wrongly diagnosed as having lumbar disc.
3. *Symptoms of intra-peritoneal rupture or retro-peritoneal leak:* The classic triad of AAA rupture is sudden severe abdominal pain + a pulsatile abdominal mass + shock. Sometimes one or more of the triad are absent in a patient with ruptured A.A.A.

Investigations

1. *US:* If an AAA is clinically suspected, US is the screening test of choice to document or to rule out the presence of an aneurysm
2. *CT scan:* If repair of an AAA is decided, CT scan provides data that are important for surgery. Spiral CT is superior to the conventional axial CT; it allows three-dimensional reconstruction of the aneurysm sac.
3. *Magnetic resonance arteriogram (MRA):* is a good alternative to CT but is expensive.
4. *Arteriogram.* An arteriogram may show the aneurysm may appear smaller if the aneurysm is occupied by a thrombus, so angiography is neither used to diagnose AAA nor to assess its dimensions. Angiography is of value if the patient has been suspected to have associated occlusive disease in the iliac, renal, or mesenteric arteries to plan the simultaneous repair of the occlusive disease.

Natural History of AAA

- *Rupture* is the most frequent and serious complication of AAA. The risk of rupture is related to the *size* of the aneurysm. The 5-year rupture rates for untreated AAA of diameters 5cm or more is high, even if asymptomatic.

Treatment

Plan of management

- *Immediate surgery* for patients with the diagnosis of rupture or acute expansion (severe pain of acute onset but with no leak on CT scan).
- *Elective surgery* for (1) *symptomatic* aneurysms regardless of the size and (2) *asymptomatic* aneurysms 5 cm or more in diameter.
- *Follow-up* is indicated every 6 months for patients with asymptomatic aneurysms <5 cm in diameter by US. Aneurysm repair is required if the serial studies show an enlarging aneurysm.

Type of surgery for aneurysm repair

1. The standard is *conventional open surgery*
 - An inlay synthetic tube or bifurcated graft is performed after proximal and distal temporary control and opening of the aneurysmal sac (Figure 21).
2. *Endo-vascular repair* of AAA (EVAR)
 - Insertion of *endo-luminal stented graft* via bilateral femoral arteriotomy (Fig. 22).

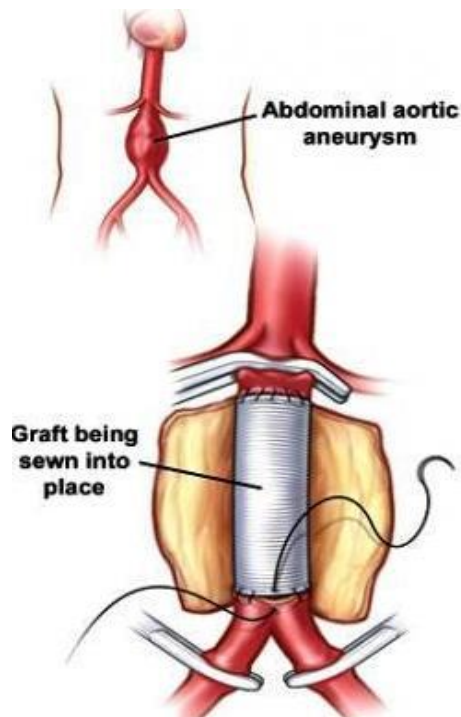


Figure 21. conventional open surgery for repair of AAA

- This technique is expensive and recurrence of the aneurysm may occur.
- It is *indicated* in:
 1. High risk patients who cannot tolerate anesthesia and open surgery.
 2. High-risk local abdominal factors such as previous major surgery or huge hostile abdomen.

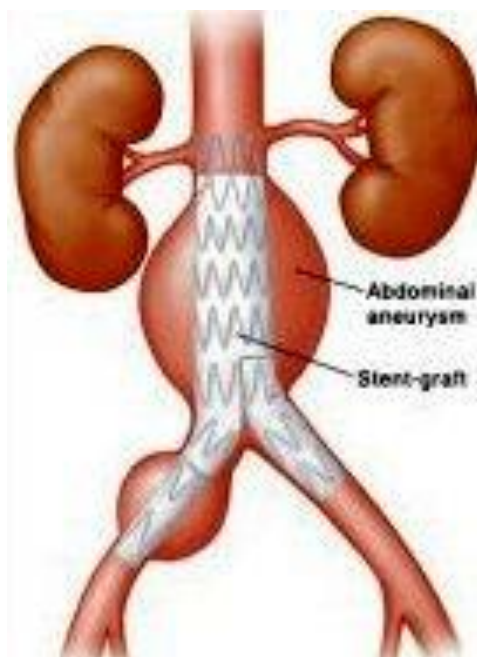


Figure 22. Conventional open surgery for repair of abdominal aortic aneurysm (AAA)

Special Difficulties

1. *Inflammatory AAA*: Avoid excessive dissection (excessive adhesions)
2. *Infected*: ligate proximal and distal to the aneurysm + debridement and extra-anatomical bypass (e.g. bilateral axillo-femoral bypass).
3. *Supra-renal*: Cool the kidney and re-implant the celiac, SMA, right renal cuff and the renal artery separately. *Fistula with IVC* (ligate from within the artery) or duodenum.
4. *Horse- shoe kidney*: If the *isthmus* is *thick*, a tunnel can be done to allow passage of the graft, but if it is *thin*, it is easy to separate it.
5. *Assessment of abdominal diseases* (e.g. cancer colon or chronic calcular cholecystitis): Do *not* open the GIT for fear of contamination and treat the most serious situation first (impending rupture obstructing cancer colon).

Post-operative Complications (General)

- Release of accumulated metabolites of the lower limb leads to vasodilatation and reduced blood pressure (intra-operatively) and causes renal shut-down (post-operatively).
- Occlusion of the graft either immediate (due to technical fault) or late (due to thrombosis).
- Graft infection (ligate, remove and extra anatomical bypass).
- Injury of ureter.
- Material failure.
- Sexual disturbances.
- Ischemia of the colon, spinal cord or limbs (distal embolization).
- Duodeno-aortic fistula, suspected by melena or hematemesis (to avoid this complication, an in-lay graft is used).

CHAPTER

5

Diabetic Foot Infection

INCIDENCE

- The number of people with Type-II DM is increasing due to change in life style and aging of the population. It is estimated to be 24% in UAE and 11% in Egypt.
- Diabetes causes more than 70% of lower limb amputations
- Feet of many of those patients are at risk of ulceration and amputation.
- Approximately, 25% of Diabetics have a diabetic foot ulcer (DFU)
- DFU precedes 85% of amputations, a large % of which could be prevented.
- Incidence of peripheral arterial disease (PAD) in diabetics is 38%. However, only 7% are assessed for arterial insufficiency.

ANATOMICAL CONSIDERATIONS

- Dorsum and sole of the foot
- Muscles “Intrinsic and extrinsic” and long tendons of the muscles of the leg
- Bones of the foot
- Arches of the foot
- Nerves of the foot
- Arteries of the foot
- Functions of the foot

Sole of the Foot

- It has a thick layer of the skin with abundant sweat glands & a thick pad of fat.

Bones of the Foot (Figure 1)

- *Tibia and fibula* (They are bones of the lower leg act on the foot)
- *Tarsus* (7): Talus, calcaneus, cuneiformes (3), cuboid and navicular (Figure 4)
- *Metatarsus* (5): 1st, 2nd, 3rd, 4th, and 5th metatarsal bones.
- *Phalanges* (14)
- There can be many *sesamoid bones* near the metatarso-phalangeal (MP) joints, although they are only regularly present in the distal portion of the first metatarsal bone.



Figure 1. bones of the lower leg and foot

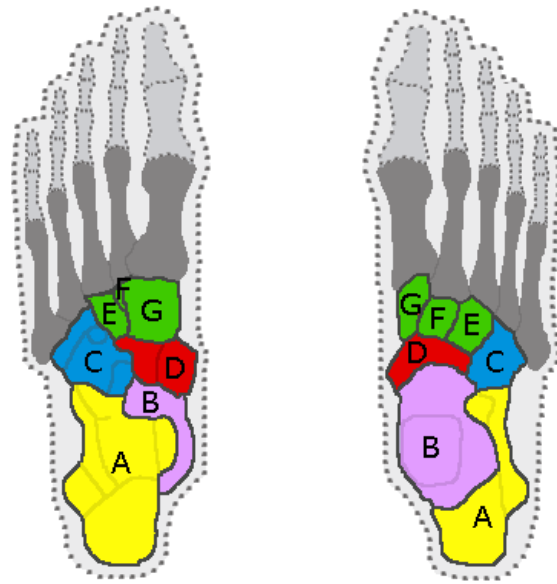


Figure 2. Bones of the right foot seen from below (left) and seen from above (right): 7 bones constitute the tarsus: **A** - Calcaneus, **B** - Talus, **C** - Cuboid, **D** - Navicular, **E, F, G** - Cuneiform bones

Arches of the Foot (Figure 3)

- The human foot has 3 *arches*, two *longitudinal arches* (one medial and one lateral) and a *transverse arch* maintained by the interlocking shapes of the foot *bones*, strong *ligaments* and pulling *muscles and tendons* during activity.
- The *medial longitudinal arch* is the highest and most important of the 3 arches. It is composed of the calcaneus, talus, navicular, cuneiforms and the first 3 metatarsals.
- The *lateral longitudinal arch* is lower & flatter than the medial arch. It is composed of the calcaneus, cuboid and the 4th and 5th metatarsals. With the cuboid serving as its *keystone*, it redistributes part of the weight to the calcaneus & the distal end of the 5th metatarsal.
- The *transverse arch* is composed of the cuneiforms, the cuboid and the 5 metatarsal bases. The 2 longitudinal arches serve as pillars for the transverse arch, which run obliquely across the tarso-metatarsal joints.
- Excessive strain on feet tendons and ligaments can cause fallen arches or *flat feet*.



Figure 3. Arches of the foot (2 longitudinal & one transverse).

Muscles of the Foot

- *Intrinsic muscles* are located *within* the foot and cause movement of the toes. These muscles are flexors (plantar flexors), extensors (dorsi-flexors), abductors and adductors of the toes. Several intrinsic muscles also help support the arches of the foot.
- *Extrinsic muscles* are located *outside* the foot, in the lower limb (LL). The powerful gastrocnemius muscle (calf) is among them. They have long tendons that cross the ankle, to attach to the bones of the foot and assist in movement. The talus, however, has no tendon attachments.

Nerves of the Foot (Sensory - Motor - Autonomic)

- Nerves of the foot help move the body and keep balance during moving and at rest. The *medial and lateral plantar nerves* are the 2 largest nerves in the bottom of the foot.
- Autonomic nerve supply controls blood flow regulation.

Arterial Supply of the Foot (Figure 4)

1. Dorsalis pedis artery: It begins as a continuation of the anterior tibial artery as it enters the foot.
2. Anterior tibial artery
3. Peroneal artery
4. Posterior tibial artery

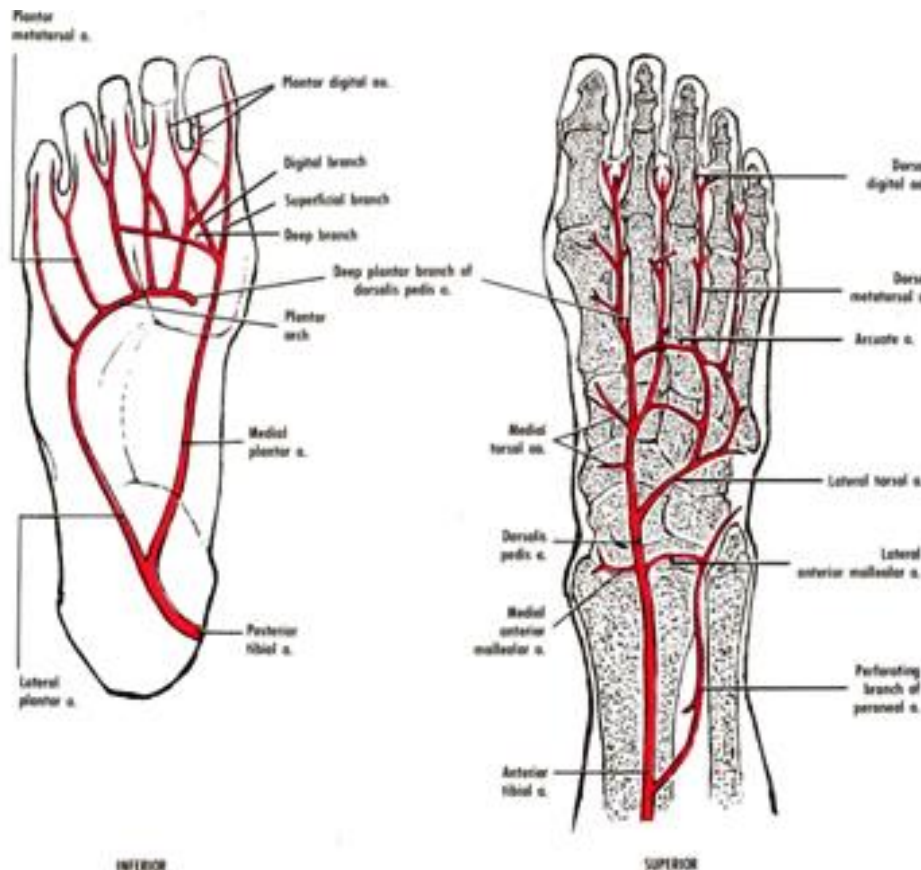


Figure 4. Arterial supply of the foot

FUNCTIONS OF THE FOOT

- The foot has the following functions:
 1. Support (weight-bearing): (5 miles /day = 1000 ton pressure /day)
 2. Movement
 3. Absorption of force
 4. Leverage action
 5. Adjustment to uneven surfaces.

DEFINITIONS

Diabetic Foot

Compilation of signs and symptoms and complications occur in the feet of patients with poor glycemic control

Diabetic Foot Infection

Any infection involving the foot of a person with DM in a chronic or acute injury to the soft tissues of the foot with or without neuropathy and /or ischemia

Charcot's Joint

Subluxation, dislocation, osteoporosis of the small bones and joints of the foot due to laxity of the ligaments and hyperemia of the bones of the foot

PREDISPOSING FACTORS

Diabetic patients are susceptible to serious foot infections due to:

1. *Peripheral neuropathy (PN)* due to axonopathy and focal compressive neuropathy. It results in reduced planter sensation + intrinsic muscle atrophy causing tendon imbalance, which leads to the following:
 - Exposure of metatarsal heads to trauma
 - Shifting weight to unprotected area leading to tissue injuries and break down of bone
 - Unawareness of foot injuries due to diminished sensation
2. *Angiopathy*
 - Premature atherosclerosis
 - Micro-angiopathy.
3. *Compromised immune system*: Both, humoral and cellular immunity are disturbed in diabetic patients resulting in inefficient functions of leucocytes (chemotaxis and phagocytosis).
4. *Hyperglycemia* that causes the following:
 - Increased capillary permeability and premature atherosclerosis
 - Increased intra-cellular *fructose and sorbitol* resulting in increased loss of tissue after mild infection and diminished immunity and resistance to infection
 - Reduced mitochondrial pyruvate utilization resulting in diminished energy.

5. *Co-morbidities*: renal, cardiac, etc.

- Poor glycemic control means HbA1c > 7%
- Each 1% increase in HbA1c leads to 15% increase in PAD
- Each 1% increase in HbA1c leads to 50% increase in PN
- Diabetic foot infection “DFI” is an emergency so that it is called “foot attack” as similar to “heart attack”

PATHOPHYSIOLOGY OF DIABETIC FOOT

Arterial insufficiency

1. Macro-angiopathy

- Aorto-iliac (Figure 7)
- Femoro-popliteal
- Tibial (infra-genicular) (Figure 8)

2. Micro-angiopathy



Figure 7. Aorto-iliac peripheral vascular disease



Figure 8. Tibial (infra-genicular) peripheral vascular disease

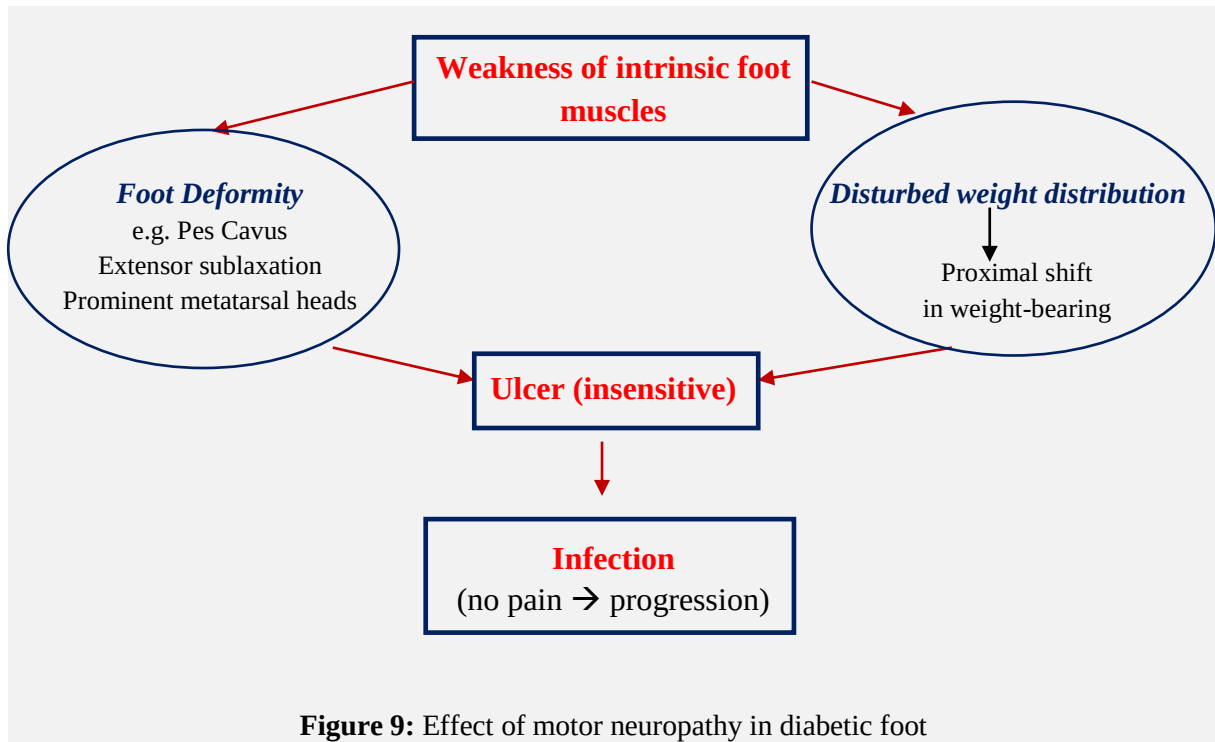
Peripheral Neuropathy (PN)

1. Sensory

- Paresthesia: Tingling, numbness, etc.
- Loss of vibration, light touch and pain
- Loss of deep tendon reflexes

2. Motor

- As shown in Figure 9, weakness of intrinsic foot muscles leads to foot deformities (Figure 10) and disturbed weight distribution
- Both conditions may lead to the development of diabetic foot ulcers (DFUs) that may become infected.



3. Autonomic (Auto-sympathectomy)

- Dry skin → fissures → infection
- Impaired vascular response to infection
- Opened AVF → Bone degeneration (Charcot's joint)

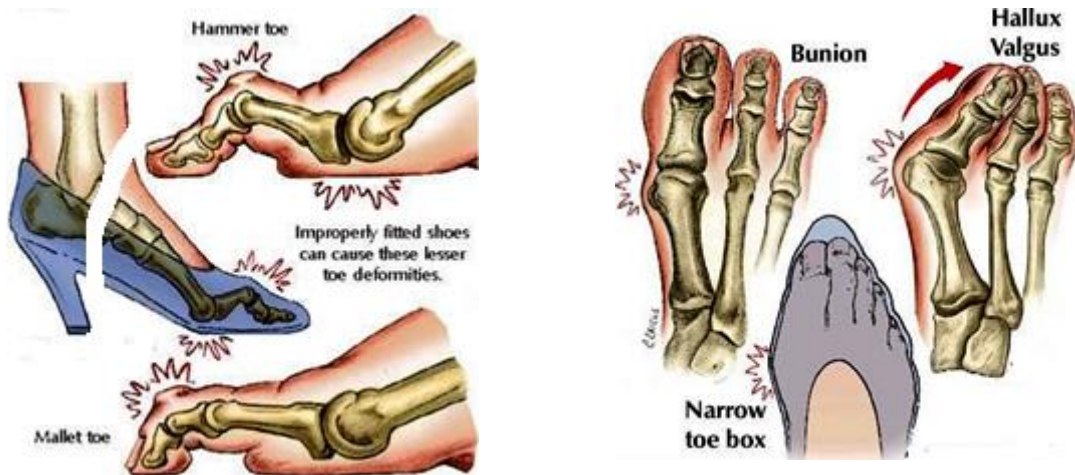


Figure 10: Pathophysiology of diabetic foot: development of diabetic foot deformities

CLASSIFICATION OF DIABETIC FOOT LESIONS

1. Wagner classification (Figure 11)
2. Brodsky classification (Figure 12)
3. WIFI classification (Wound, Ischemia, Foot Infection) (Figure 13)



Figure 11: Wagner classification of diabetic foot ulcers (DFUs): five grades (1-5)

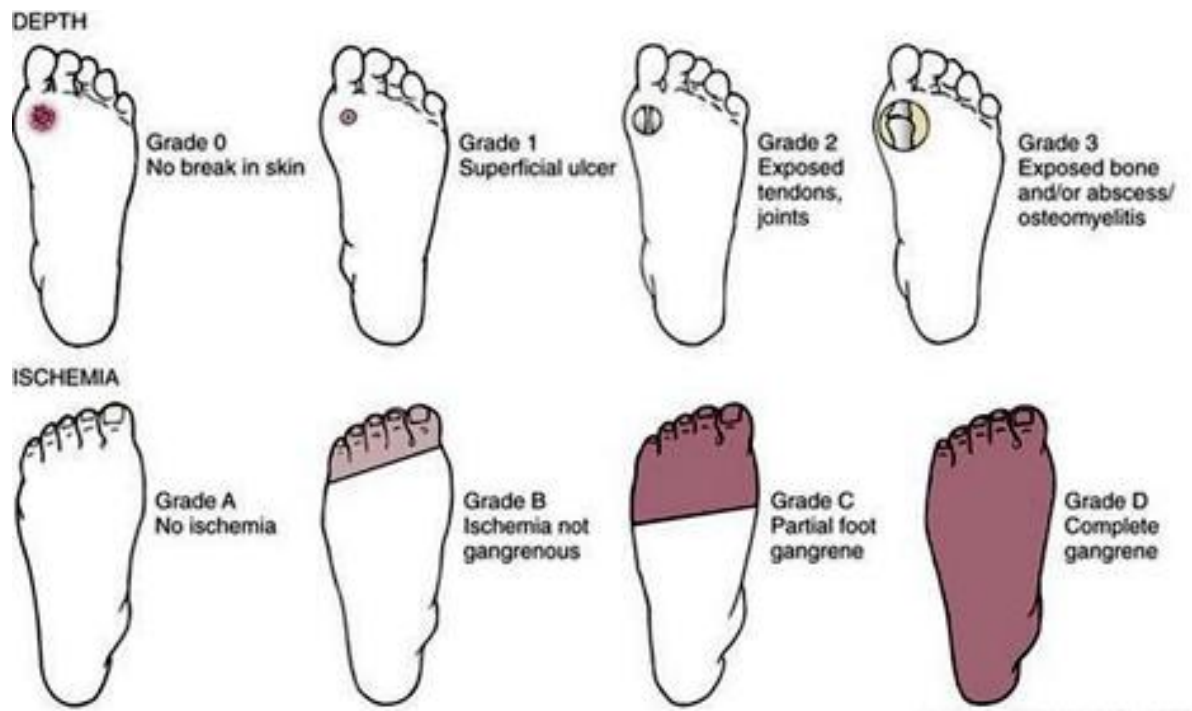


Figure 12: Brodsky classification of diabetic foot ulcers based on depth and ischemia

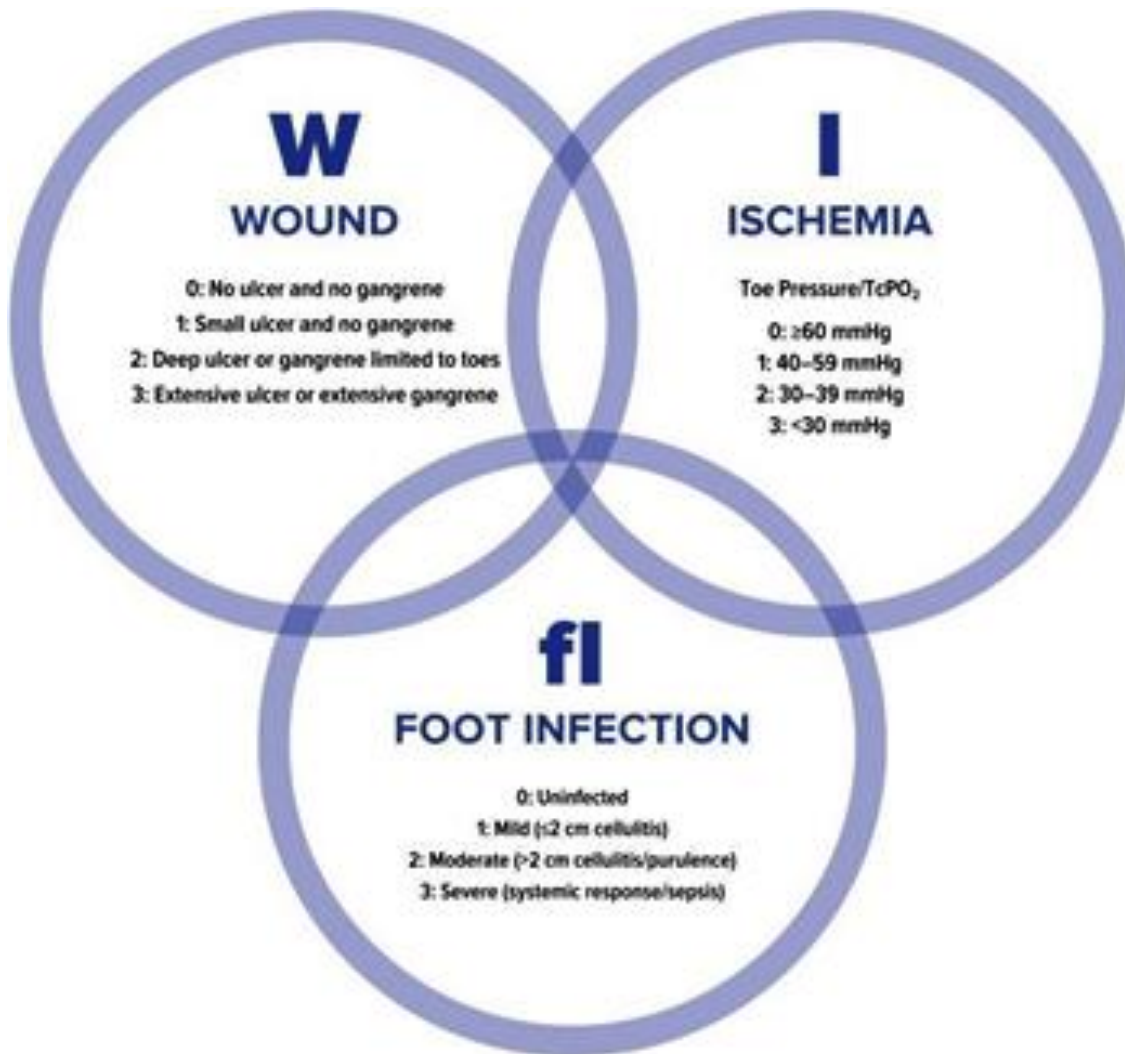


Figure 13: WIFI classification of diabetic foot ulcers based on wound, ischemia and foot infection

CLINICAL PRESENTATION

- Most of serious diabetic foot infections start by a trivial *trauma* or *infection*, eg a pin prick, careless cutting of a nail, scraping of a callosity or inter-digital fungal infection. Initial injury or infection is usually overlooked by the patient.
- Consequences could be involvement of soft tissues, bones and joints.

Types of Diabetic Foot Lesions

1. Neuropathic ulcer (Figure 14).
2. Ischemic ulcer (Figure 15).
3. Locally infected ulcer (Figure 16).
4. Mixed lesions

Clinical Presentation of Ischemic Lesions

1. Rest pain (43%).
2. Toe gangrene (33%) (Figure 17)
3. Unhealed foot amputation stump and neuro-ischemic ulcers (24%)



Figure 14. Neuropathic ulcer in a diabetic patient with Charcot joint.



Figure 15. Ischemic ulcer on the lateral aspect of the left foot of a diabetic patient.



Figure 16 An infected diabetic ulcer. Note the infected, gangrenous areas and cellulitis.



Figure 17. Gangrene of the big toe of the left foot. Note the severe cellulitis

Infective Lesions and Ulcers

Types

1. Superficial skin infections
2. Deep infection or ulcer involving tendons, ligaments or bones
3. Deep infection or ulcer with osteomyelitis or septic arthritis
4. Infective gangrene of the forefoot (Figure 18) or heel (Figure 19)
5. Infective gangrene of the whole foot (Figure 20)

Severity

1. Non-threatening
2. Threatening with moderate infection
3. Threatening with severe infection and toxemia

Bones and Joints

- Osteomyelitis (with deep infection)
- Septic arthritis (with deep infection)



Figure 18. Gangrene of forefoot



Figure 19. Gangrene of forefoot and heel



Figure 20. Gangrene of the whole foot

Complications

- *Spread of infection:* Necrotizing fasciitis involves fascial spaces and tendon sheaths. Multiple pockets of pus develop and the tissues become necrotic with grey or black sloughs. The patient looks toxic with high pyrexia. Unless the condition is treated as an emergency, infection may spread to involve the leg or even the thigh.
- *Septicemia and septic shock* may develop if pus is not drained.
- *Diabetic gangrene* of the feet may be due to one of the following:
 1. *Pure infection.* There is no major vascular affection and all the pulses and sensations are intact (wet gangrene).
 2. *Pure ischemia.* This is a (dry ischemic gangrene) occurring in a diabetic patient with major vascular obstruction.
 3. *Mixed type.* There is severe infection + major vascular and neuropathic affection.

Physical Examination – Assessing Diabetic Foot

Assessment of the diabetic foot entails 8 factors (Table).

Evaluation of Neuropathy

- Up to 35% of all diabetic patients have asymptomatic neuropathy and in these patients it will only be detected by examination
- The presentation of peripheral neuropathy (PN) is related to dysfunction of sensory, motor and autonomic nerves.

Motor Neuropathy

- The classical sign of motor neuropathy is a high medial longitudinal arch leading to prominent metatarsal heads and pressure points over the plantar forefoot (Figure 21).
- It is advisable to test dorsi-flexion of the foot to detect a foot drop 2nd to a common perineal nerve palsy which is usually unilateral & will affect the patient's *gait*.

Autonomic Neuropathy

- There is dry skin with fissuring secondary to decreased sweating which normally occurs in a stocking distribution above which there is excessive sweating.
- Distended veins are seen over the dorsum of the foot and ankle secondary to arterio-venous shunting associated with autonomic neuropathy.

Sensory Neuropathy

- Sensory neuropathy can be detected by:
 1. Nylon monofilaments: The filament should be pressed against the plantar aspects of the 1st toe, 1st, 3rd and 5th metatarsal heads, plantar surface of the heel, and dorsum of the foot (Figure 22). If the patient cannot deal with the filament at any of the tested areas, then significant neuropathy is present and protective pain sensation is lost.
 2. For detecting sensation to light touch, a cotton wisp is used and for vibration, a 128-Hz tuning fork is used, comparing a proximal site with a distal site.
 3. A simple temperature assessment of cold sensation can be made by placing a cold tuning fork on the patient's foot and leg.
- Patients with small-fiber neuropathy have impaired pain and temperature perception but with intact touch and vibration. These patients are prone to ulceration and thermal trauma.

Clinical Assessment of the Diabetic Foot

Neuropathy
Ischemia
Deformity
Callus
Swelling
Ulceration
Infection
Necrosis / gangrene



Figure 21. High arched foot with prominent metatarsal heads



Figure 22. Nylon monofilament testing of peripheral neuropathy

Evaluation of Arterial Insufficiency (Ischemia)

- Ischemia results from *arteriosclerosis* of the leg. Claudication is often absent due to concomitant neuropathy. The *skin* is thin, shiny and hairless, with atrophy of SC tissue. Skin *color* is often dusky red or cyanotic due to impaired perfusion, with stagnation of blood in dilated arterioles. In *acute ischemia*, the foot is pale, often with purplish mottling.
- The important clinical maneuvers to detect ischemia are:
 1. **Palpation of the foot pulses;** dorsalis pedis and tibialis posterior (Figure 23): If either pulse can be felt, then it is unlikely that there is significant ischemia.
 - Dorsalis pedis pulse is found lateral to the extensor hallucis longus (EHL) tendon on the dorsum of the foot.
 - Posterior tibial pulse is palpated above and behind the medial malleolus.
 - Femoral pulse is palpated as it enters the femoral triangle at the *mid -inguinal point*.
 - The popliteal artery is the hardest pulse to find. It lies deep in the popliteal fossa and requires deep palpation to feel. To make it easier, you can ask the patient to slightly flex their leg – this relaxes the fascia around the popliteal fossa.



Figure 23: Palpation of arterial pulses; A: dorsalis pedis, B: posterior tibial and C: popliteal arteries

2. Hand-held Doppler

A small hand-held Doppler (Figure 24) can be used to confirm the presence of pulses & to quantify the vascular supply.

3. Ankle/Brachial Index (ABI)

Used together with a sphygmomanometer, the pressure index (ABI) which is *normally* >1 can be measured (Figure 25). Absence of pulses & an ABI < 1 confirm ischemia. If the ABI is >1 but pulses are not palpable, then ischemia probably exists in the presence of medial wall calcification. In such cases, other methods to assess flow in the arteries

of the foot should be used such as (1) examining the pattern of the *Doppler arterial waveform*, or (2) measuring *trans-cutaneous O₂ tension* or (3) *toe systolic pressure*.



Figure 24. Hand-held Doppler examination of the posterior tibial

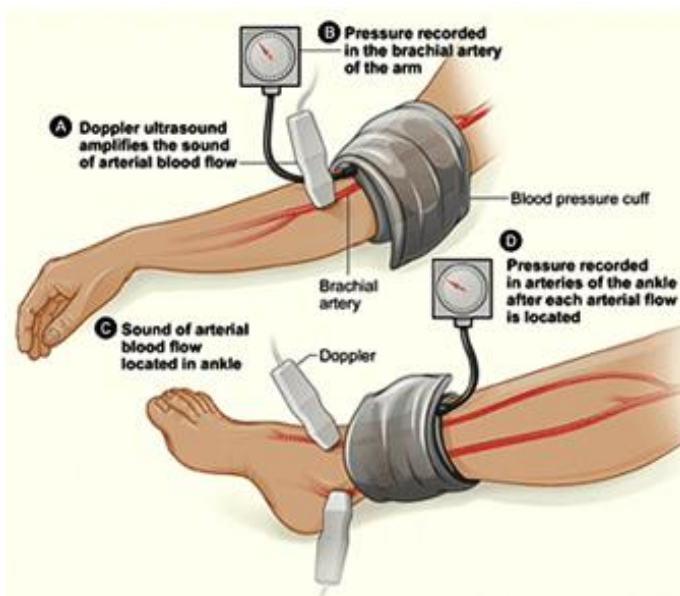


Figure 25. Measurement of the ankle / brachial index (ABI)

Investigations

1. Laboratory

- Blood glucose level, HbA1C (normal < 7), each 1% ↑ → 15% ↑ in PAD and 50% ↑ in peripheral neuropathy)
- Serum levels of lipids (Cholesterol & triglycerides).
- *Culture and sensitivity (C&S)* from pus.

2. ECG

3. Imaging of the Foot

- *Plain X-ray*
 - Bone (osteomyelitis, necrosis, fractures, dislocations) – Figure 26.
 - Medial arterial calcification
 - Soft tissue gas (*Clostridial infection*, gas-forming organisms, or due to milking of air during walking).
 - Charcot joint.
- *CT scan*: It delineates suspected bone or joint pathology not evident on plain X-ray.
- *Radioisotope Technetium bone scan*: It can detect early pathology such as osteomyelitis, fractures and Charcot arthropathy.
- *MRI*: It allows evaluation of both soft tissue and bone pathologies.



Figure 26. Plain X-ray of the foot. Note osteomyelitis (big toe)

4. Evaluation of Arterial Insufficiency

- *Clinical Examination “Pulse”*: Refer back
- *Ankle/Brachial Index “ABI”*: Refer back
- *Toe/Brachial Index*: Accurate measurement of toe pressures requires sensitive techniques such as *laser Doppler*.
- *CDU*: Colored Duplex Ultrasound Scan
- *CTA*: CT angiography

Management of Diabetic Foot

A. Preventive Measures

Diabetic patients should be advised to do the following:

- *Proper control of DM* by diet and by medications such that HbA1C is < 6.
- *Avoid risk factors* e.g. stop smoking.
- *Daily foot care*
 1. Washing, drying (between the toes) and powdering.
 2. Daily inspection of the feet to look for redness, induration, hotness, blisters, cracks, fissures, wounds and inter-digital infections. Peripheral neuropathy allows such lesions to pass unnoticed till infection sets in.
- *Regular foot care*
 1. Careful trimming of toe nails to avoid their injury.
 2. Early treatment of *tenia pedis* (fungal) infection.
- *Foot wear*: Avoidance of tight foot wear or walking bare-foot.
- *Prevent foot injury by avoiding* walking bare-feet, hot water, sun bath, ill-fitting shoes and hard objects.
- *Regular medical check-ups* (ABI, hand-held Doppler, neuropathy, etc)

B. Treatment of Early Lesions

- Detection and treatment of early lesions (Figure 27)
- Special centers “Podiatrist” → foot care and management of early lesions
- Smart phones can also be used in diabetic foot for early care and management



Figure 27. Clinical examples of early diabetic foot lesions (small ulcers and minor gangrene)

C. Proper Intervention of Foot Lesions

1. Hospitalization.
2. Rest in bed and elevation of the foot.
3. Blood glucose is estimated for proper control of DM; better by insulin every 8 hours.
4. Neuropathic ulcers
 - a) Foot print
 - b) Off-loading (Figure 28)
 - c) Management of pes performans (open joint): Dorsal excision of MPJ → removal of the septic focus + elimination of pressure points
5. Infective lesions and ulcers
 - a) Broad spectrum antibiotics are started immediately until the results of culture are available. In many patients, anaerobic organisms are involved and should be covered. The dose of the antibiotic and duration of treatment should be adequate
 - b) Extensive debridement for any necrotic tissues
 - c) Incision and drainage of any collection: Under general anesthesia, an experienced surgeon should widely drain all pockets of pus. All sloughs should be excised. A gangrenous toe needs to be *amputated* leaving the wound open.
 - d) Repeated dressings and repeated debridement with drainage of pockets of pus if present. The use of VAC therapy (Figure 29) enhances wound healing.



Figure 28. Off-loading to enhance healing of early ulcers and avoid their recurrence.

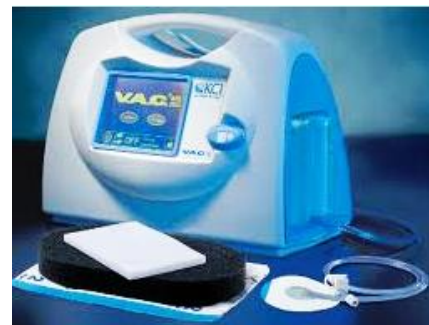


Figure 29. VAC machine used in management of raw area to enhance wound healing

- e) When the wound becomes completely free of infection and contains only viable tissue, a large raw area is considered for *plastic skin coverage*.
- f) In some patients, infection is widespread and major amputation is necessary to save patient's life.

6. *Ischemic ulcers and arterial insufficiency*

- *Medical treatment*
 - Cilostazol, Naftidufuryl, Pentoxifylline
 - Antiplatelets
 - LMWH (low molecular weight heparin)
- Patients with occlusion of a major artery should have an angiography to assess the feasibility of *arterial reconstruction*.
- *Bypass surgery e.g. femoro-popliteal* (Figure 30)
- *Endo-vascular intervention (refer back)*

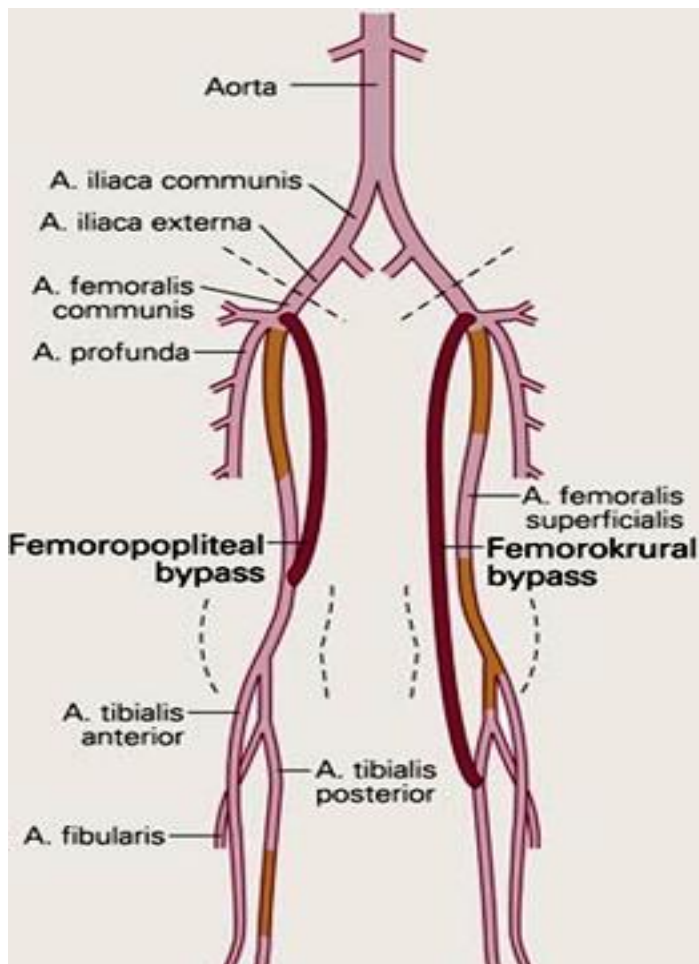


Figure 30. Bypass surgery for management of arterial occlusive disease

7. *Charcot foot*

- *Acute* (bed rest, Bisphosphnate, off-loading)
- *Chronic*: Off-loading "Air walker".

Charcot Foot

Definition

- Subluxation, dislocation, osteoporosis of the small bones and joints of the foot due to laxity of the ligaments and hyperemia of the bones of the foot (named after a French surgeon).

Incidence

- It occurs in approximately 10% of patients with diabetic peripheral neuropathy (PN) and 16% of patients with neuropathic ulcer

Etiology and Pathophysiology

- Sympathetic neuropathy and hyperemia → decalcification of the bones and then fractures or subluxation of the joints.
- Sensory PN → neuropathic ulcer and weight bearing (Figure 31)



Figure 31. Charcot foot: Note the presence of the neuropathic ulcer

Stages

1. Acute stage: Swelling (Figure 32) and hotness
2. Subacute stage: Loss of foot arch “flat foot” (Figure 33)
3. Chronic stage: Fractures and dislocation (Figure 34)



Figure 32. Charcot foot: Acute stage with foot swelling



Figure 33. Charcot foot: Subacute stage with *flat foot*

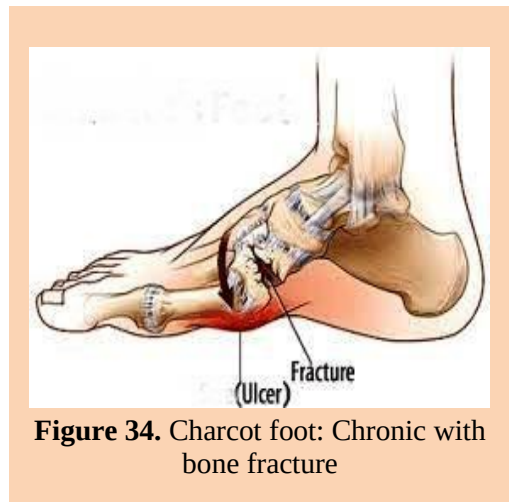


Figure 34. Charcot foot: Chronic with bone fracture

Investigations

1. Plain X-ray (Figure 35)
2. FNAB (fine-needle aspiration biopsy)
3. Foot Print (Pedobarography) (Figure 36) is the study of pressure fields acting between the plantar surface of the foot and a supporting surface.
4. MRI (magnetic resonance imaging)



Figure 35. Plain X-ray showing Charcot foot

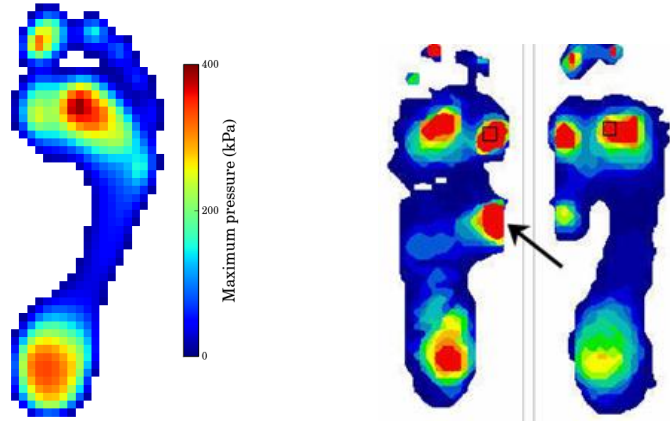


Figure 36. Foot print showing the pressure fields on the plantar surface (*Left*). Note the high pressure in Charcot foot (red color-arrow) (*Right*).

Management

1. *Acute Stage*
 - Bed rest
 - Bisphosphnate
 - Total Contact Cast and off-loading.
2. *Chronic Stage*
 - Off-loading “air walker”

CHAPTER

6

Vascular Trauma

Magnitude

- A major health problem as it is the main cause of death before the age of 38 years.
- Less than 10% of patients with multiple-trauma but with serious outcome in terms of morbidity and mortality.
- Different types of vascular injuries among different societies.

Etiology (Types) According to Mechanism of Injury

- *Blunt trauma*: road traffic accidents (RTAs), falling from a height, direct hit.
- *Penetrating injury*: stab, low-velocity missile, high-velocity missile, slash injury
- *Iatrogenic injuries*: more than 40%.

Pathophysiology

Blunt Trauma

- The weakest layer of the vessel wall is the tunica intima followed by tunica media and adventitia. So, arterial contusion can produce intimal disruption and occlusion with artery in continuity. But, when force of blunt injury is high, complete vessel disruption could happen.
- Blunt vascular injury includes:
 1. Direct trauma to the vessel wall: accounts for the majority of cases. Producing vessel wall contusion with intramural haematoma, intimal disruption, intimal flap and luminal thrombosis.
 2. Indirect injury: less common, could happen due to shearing or distraction force applied to the vessel wall either by major joint dislocation or displaced long bone fractures as well as acceleration/deceleration injuries.
- This will lead to immediate occlusion, delayed occlusion, complete disruption.

Penetrating Injury

- The result is arterial cut either complete or incomplete.
- Stab or low velocity missile.
- High velocity missile injury: cavitation injury and shock wave effect.

Sequelae of Vascular Injuries

1. Bleeding: acute, delayed, rebleeding.
2. Thrombosis: acute, delayed, ischaemic manifestations.
3. Haematoma with possible infection.
4. Pseudo-aneurysm formation.
5. Arterio-venous fistula.

Clinical Picture

History-Taking

- History of trauma and its mechanism
- Bleeding and its description
- Previous underlying vascular disease.

Clinical Examination (Signs of Vascular Trauma)

- *Hard signs of vascular injury*
 1. Active pulsatile bleeding.
 2. Shock with on-going bleeding.
 3. Pulsatile or expanding haematoma.
 4. Absent distal pulse.
 5. Manifestations of acute ischemia.
 6. Palpable thrill over the site of injury.
- *Soft signs of vascular injury*
 1. History of marked bleeding.
 2. Diminished distal pulses.
 3. Small nonexpanding haematoma.
 4. Injury of anatomically nearby structures
 5. Injury at the anatomical site of major vessels.
 6. Multiple fractures and extensive soft tissue injury.

Investigations

Plain X-ray

- It raises the index of suspicion when there is a complex fracture or fractures at certain anatomical sites related to major blood vessels.

Duplex Ultrasound

- It has important roles in:
 1. Extremity vascular trauma.
 2. Neck zone-2 trauma (Figure 1).
 3. Abdomen (Figure 2).
- It is of value in both, screening of patients with soft signs, as well as follow-up of patients treated expectantly.

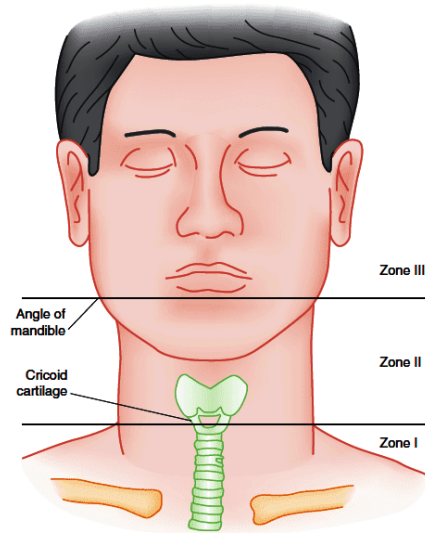


Figure 1: Zones of cervical vascular injuries

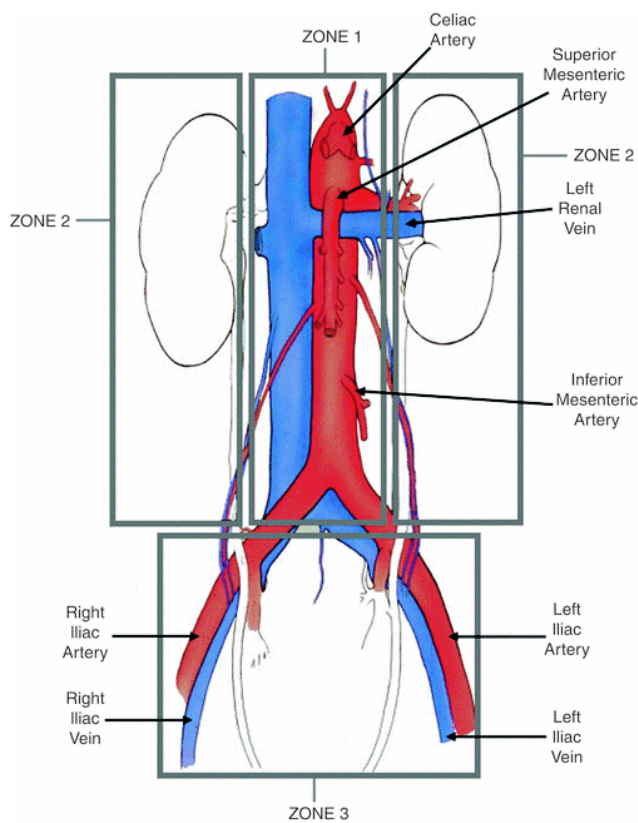


Figure 2: Zones of abdominal vascular injuries

Arteriography

- It is indicated in stable patients, in the presence of hard signs, in cases of:
 1. Multiple complex fractures.
 2. Extensive soft tissue injury.
 3. Gunshot injuries.
 4. Neck injuries zones 1 and 3.
 5. Thoracic and abdominal injuries.

Treatment

Resuscitation and First-Aid Management

- Resuscitation: following ATLS (Advanced Trauma Life Support) guidelines, of most importance.
- Investigations should not delay intervention especially in unstable patient.
- The use of permissive hypotension by avoiding aggressive fluid resuscitation in hypotensive patients when bleeding control is not done (hypotensive resuscitation).
- The use of digital compression, and tourniquet to stop bleeding temporarily (blind clamping is absolutely prohibited).

General Principles of Surgical Intervention

- General anesthesia.
- Antibiotic prophylaxis.
- Well-equipped theater
- Blood transfusion ready to give on indication.
- Preparation of necessary areas for proper exposure.
- Proper exposure for adequate proximal and distal vessel control before exposure of site of injury.
- The use of digital pressure.

Venous Injuries

1. Simple small cut: direct repair.
2. Large cut with or without laceration: we do either trimming with end-to-end anastomosis, or interposition vein graft repair.
3. In cases of massive laceration, long segment loss, or a sacrificable vein, ligation could be done.

Arterial Injuries

- Trimming of the arterial cut and excision of the contused segment until we get a healthy tunica intima then:
 1. Direct lateral repair when we have partially cut artery affecting less than 50% of the wall circumference.
 2. Direct end –to-end anastomosis when partial cut is larger than 50%, completely cut artery, or a short contused arterial segment is found.
 3. Interposition graft bypass to repair long arterial contusion or lacerations affecting long segment that cannot be directly repaired (reversed vein graft is always preferred).

In Damage Control Surgery

- When we have critically ill patient with multiple trauma who is risky for definitive repair, we do either ligation or shunting of the injured vessel as a temporary manoeuvre until correction of the patient general condition preparing him for definitive surgical intervention.

Surgical Treatment of the Sequelae of Vascular Injuries

1. Surgical treatment of chronic post-traumatic limb ischemia e.g. arterial bypass grafting
2. Separation and repair of traumatic arterio-venous fistula.
3. Repair of traumatic aneurysm and pseudo-aneurysm.
4. Drainage of non-communicating haematoma.
5. Amputation of limbs with post traumatic gangrene and irreversible tissue damage.

Role of Endovascular Procedures in Vascular Trauma

- *In the acute phase*
 1. Stoppage of active bleeding by coiling of a feeding artery or bunch of veins.
 2. Preliminary control of bleeding until definitive repair using trans-luminal occlusion balloons.
 3. Repair using stent grafts in thoracic aorta, thoracic outlet vessels, distal internal carotid and vertebral arteries where exposure is difficult and time consuming.
- *For treatment of vascular trauma sequelae*
 1. Arterial stenosis following arterial injuries
 2. Pseudo-aneurysms following arterial injuries.

Venous Disorders

SURGICAL ANATOMY AND PHYSIOLOGY OF LOWER LIMB VEINS

Venous Drainage of the LL (Figure 1)

1. *Superficial system*: Long saphenous vein (LSV) and short saphenous vein (SSV)
2. *Deep system*
 - 3 pairs (anterior, posterior tibial and peroneal) of venae comitantes form the popliteal vein
 - Gastrocnemial veins: Two veins for each head of the muscle
 - Soleal (sinusoids)
 - posterior tibial or peroneal venae comitants
3. *Perforators & communicators*

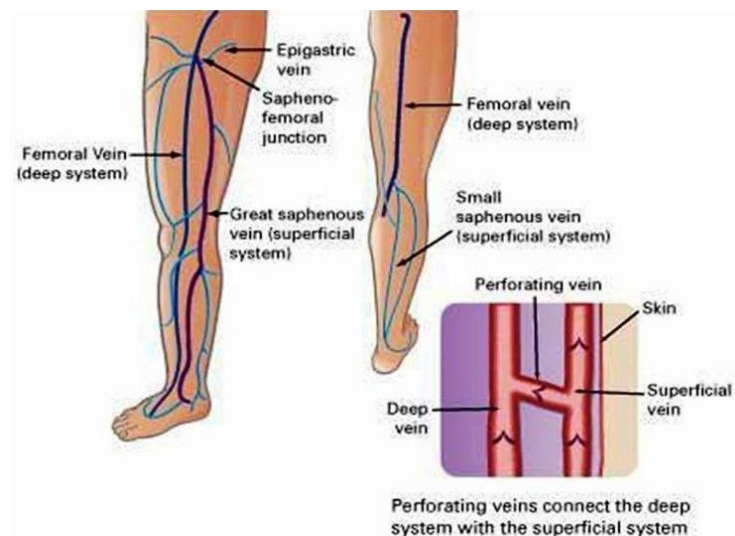


Figure 1. Venous anatomy of the lower limb (LL)

Superficial Veins

These are SC veins that lie *superficial* to deep fascia (DF).

Long Saphenous Vein (LSV)

- *Beginning*: This vein begins as an upward continuation of the medial marginal vein of the dorsal venous arch of the foot.
- *Course and relations*: The vein runs upwards anterior to the medial malleolus. It ascends along medial aspect of leg behind the medial border of the tibia towards the knee where it lies a hand breadth behind the medial border of the patella (an important surface marking), it then spirals forwards round the medial convexity of the thigh. The saphenous nerve

accompanies the long saphenous vein in the lower part of its course below the knee very closely, and then up to the groin slightly away from it.

- **Termination:** It ends by passing through the cribriform fascia covering the saphenous opening, which lies 3.5 cm below & lateral to the pubic tubercle, where it joins the antero-medial side of the femoral vein.
- **Tributaries:** Four tributaries in the region of the saphenous opening corresponding to the branches of the femoral artery; namely, (1) superficial circumflex iliac, (2) superficial epigastric, (3) superficial external pudendal and (4) deep external pudendal.
 - The postero-medial and antero-lateral veins of the thigh drain a wide area of the thigh and join the LSV below the saphenous opening. The supra-patellar vein draining the anterior surface of the knee.
 - The anterior (superficial tibial vein) in the leg and posterior arch veins of the leg, which drain a wide area of the leg. The posterior arch of **Leonardo's** in the leg is connected to the deep system by **Cockets's** perforators.

Short Saphenous Vein (SSV)

- It drains the lateral side of the dorsal venous arch & the lateral margin of the foot.
- It lies with the sural nerve *behind* the lateral malleolus, then passes upwards in the SC fat along the midline of the calf and pierces the DF anywhere from the mid-calf to the roof of the popliteal fossa where it enters the popliteal vein.
- It communicates with the long saphenous vein by several small veins.

Perforating Veins (Figure 2)

- These are short veins that perforate the DF to connect superficial vein to deep veins.
- They are very plenty, only few of these are functioning.
- Perforators may communicate tributary of the SSV to the deep veins.
- All perforators have *uni-directional* valves to allow blood to pass only from the superficial to deep veins.
- The skin of the lower 1/3 medial aspect of the leg is additionally drained by special veins, which pierce the DF & drain directly into the deep system of veins. These veins are called "*ankle perforators*".
- **Perforating veins**
Connect superficial & deep system either directly or indirectly via muscle sinusoids. The *perforators* drain the skin & SC directly to deep system. They are the main ones affected in PPS.

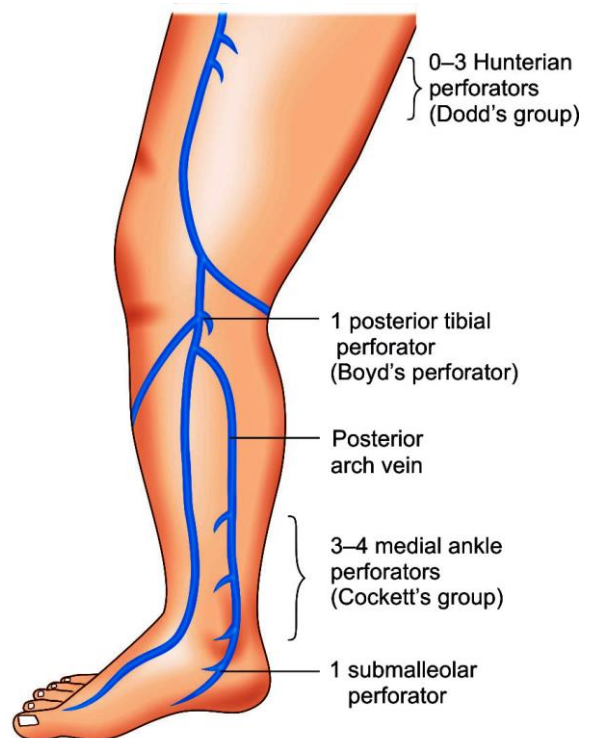


Figure 2. Perforating veins of the lower limb

- **Communicating veins**

They communicate superficial veins with either named veins or tributaries to another superficial vein i.e. between tributaries of both SSV and LSV.

- The anatomically known perforators are:

A. **With the LSV**

1. Group of perforators:

- **Cocket's I**: 6-7 cm behind the medial malleolus from the solar aspect of the foot.
- Cocket's II: 13.5 cm.
- Cocket's III: 18.5 cm.
- Perforator 24, which is 24 cm from the medial malleolus.

2. Below the knee at the tibial tuberosity (**Boyd's**).

3. Medial aspect of the thigh (sub-sartorial) (**Dodd's**).

B. **With the SSV**

1. **Bassi's**: 5 cm above the calcaneum at the lateral aspect of the leg.

2. Solus veins with the superficial system.

3. Gastrocnemius perforators.

- **NB**: There is a vein "sciatic vein" that connects both, the internal iliac vein and profunda femoris.

Venous Drainage of the UL

Superficial Veins

- Cephalic vein (lateral): It joins the axillary vein by piercing the clavipectoral fascia
- Basilic vein (medial)

Deep Veins

- Venae comitantes of the radial, ulnar and brachial arteries.
- Venae comitantes & Basilic veins, at the posterior axillary fold to form the *axillary vein*.

Factors that Help the Venous Return from the LLs

1. The muscle pump, which needs strong muscles and intact DF.
2. Transmitted arterial pulsations to the venae comitantes.
3. The negative intra-thoracic pressure.
4. The uni-directional valves. The venous system of the LL is characterized by the presence of uni-directional bicuspid valves, which allow blood to pass from below upwards and from superficial to deep veins. The venae cavae are exceptions as they have no valves.

ACUTE VENOUS THROMBOSIS

Pathogenesis

Virchow, in 1856, mentioned his **triad** for the factors predisposing to venous thrombosis.

1. *Damage to the endothelial lining of the vein wall due to:*

- Trauma to the vein wall e.g. during pelvic operations.
- Inflammatory process near the vein e.g. pelvic sepsis.

2. *Venous stasis due to:*
 - Prolonged bed rest, long trips, or casts.
 - Congestive heart failure.
 - Venous compression by tumors or a pregnant uterus.
3. *Hyper-coagulability of blood due to:*
 - Deficiency of anti-thrombin III, proteins S & C, macro-globulins or antitrypsin.
 - Polycythemia.
 - Factor V Leiden gene defect or activated protein C resistance.
 - Dys-fibrinogenemia.
 - Anti-phospholipid syndrome.

I. Superficial Vein Thrombosis (Superficial Thrombophlebitis)

Etiology

Superficial thrombophlebitis may occur in the following conditions:

1. Varicose veins.
2. Veins cannulated for IV infusion.
3. After injection of irritant drugs, e.g., diazepam.
4. In association with Buerger's disease (thrombophlebitis migrans), polycythemia, Polyarteritis and visceral cancers (it may be the earliest sign of malignancy thrombophlebitis migrans **Trousseau sign**)).
5. Idiopathic.
6. Specific Form (Mondor's disease).
7. **Behcet** syndrome

Clinical Features

- The vein becomes red, painful and cord-like.
- There may be slight pyrexia.

Complications

1. Suppuration: Treatment = Excision of the vein, drainage and antibiotics
2. Extension to the deep system via perforating veins. Treatment: Ligation of the vein above the thrombus.
3. The thrombus is adherent to the vein wall as there is inflammation. Pulmonary embolism (PE) occurs only very rarely.

Treatment

Conservative Treatment

- Compression by elastic stocking or compression bandage.
- Anti-inflammatory drugs e.g. aspirin.
- Antibiotics only if there is evidence of infection.
- Anti-coagulant therapy (Heparin and Warfarin) is given in severe progressive cases, if DVT is suspected, or if it begins at first in the thigh where there is a high possibility to extend to the deep system

Surgical Treatment

- Prophylactic sapheno-femoral or sapheno-popliteal disconnection is done if the process of thrombosis is ascending towards the junction with the deep system.

II. Deep Venous Thrombosis (DVT)

Predisposing Factors

- All the factors mentioned before under *Virchow's triad*, in addition to (1) Oral contraceptive intake, (2) obesity, (3) malignancy, (4) major trauma, (5) old age & (6) previous DVT.

Pathogenesis

- The beginning: The process usually starts in the calf venous sinuses, in the iliac or femoral veins by adherence of platelets to the endothelial surface.
- Early: When more platelets adhere, fibrin and RBCs are deposited as layers in-between the platelets giving a *laminated appearance*. When the vein is totally occluded, non-adherent, jelly-like (propagated thrombus) spreads up the vessel as far as the next major tributary. This thrombus is dark red and consists only of fibrin & red cells. This thrombus is *loosely* attached and it can be easily detached leading to PE.
- Later, the thrombus becomes tightly adherent to the vein wall by fibrin deposition.
- Sequelae: It then organizes and contracts → destruction of the valves and narrowing of the lumen, which → eventual development of the *post-phlebitic limb syndrome*.
- Resolution: Later on, the processes of fibrinolysis and phagocytosis start and help in recanalization of the vein, but the *valves are permanently destroyed*.

Clinical Features

The clinical picture of DVT varies greatly from *silent* cases to severe local and systemic symptoms.

A. DVT of the Lower Limbs

Risk Factors

- Pelvic or prostatic surgery - ♀ >40 y, using oral contraceptive pills (OCP) – Malignancy – Obesity - Prolonged recumbency - Past history of DVT.

Silent DVT

- There are *no* local symptoms and the patient may present with either PE or later with the manifestations of a post-phlebitic limb.
- This occurs when thrombosis begins in the soleal plexus of veins & extends proximally.
- It may be *non-occlusive*.

Symptomatic cases

Symptoms

1. Pain (usually aching) and tightness with sense of fullness or tightness in the involved calf or thigh, which are aggravated by muscular exercise.

2. Swelling is evidenced by measuring the difference in the circumference between both legs. The extension of the swelling depends on the site of DVT:
 - In *calf* thrombosis, the swelling is limited to the foot and ankle
 - In *femoral* thrombosis, the swelling involves the calf and lower part of the thigh,
 - In *ilio-femoral* thrombosis, there is massive swelling affecting the whole LL.
3. Pulmonary embolism manifested by severe chest pain, dyspnea or hemoptysis.

Signs

1. Color; Bluish.
2. Prominent veins at the dorsum of foot, which feels hot.
3. Ankle edema: If mild, you can detect it by resistant to pinch the skin at the ankle.
4. The calf muscle becomes stiff with ↑ muscle turgor.
5. Tenderness is present on grasping the affected calf or thigh or on compressing the muscles by the 2 hands against the bones. Pain occurs along the course of the femoral vein. *Dorsi-flexion* of the foot causes pain in the calf (**Homan's sign**). To differentiate between it & *myositis* of DM, squeezing the calf muscle from side to side gives more pain in myositis. However this sign is not reliable. A +ve **Homan's** sign denotes thrombosis of the posterior tibial vein. It is better avoided for fear of dislodgment of emboli.
6. Comparative measurement

Advanced Clinical Types

1. *Phlegmasia Alba Dolens (PAD)*

- Massive ilio-femoral DVT may be associated with severe arterial spasm.
- The limb becomes pale white, massively swollen with absent peripheral pulses & severe pain.

2. *Phlegmasia Cerulea Dolens (PCD)*

- Begins as PAD.
- Characterized by severe congestion, cyanosis & pain.
- Tingling & numbness are the 1st manifestation then bursting pain.
- ± Shock as a large amount of blood is sequestered in the limb.
- Mostly pulsations cannot be felt

3. *Venous gangrene*

- It occurs in advanced, extensive, neglected DVT.

B. Thrombosis of Axillary Vein

- Either spontaneously or following trauma.
- Superficial veins are more prominent.
- Edema of the UL.
- The axillary vein is cord-like.

C. Thrombosis of SVC

- Dilated veins over chest
- Abdominal edema
- Plethora of the H&N.

Investigations

Laboratory

- ↑ ESR, hyper-lipidemia and ↑ triglycerides (TG).
- Protein S and C deficiency and anti-thrombin III deficiency when a hereditary cause is suspected
- Tumor marker if malignancy is suspected.

Imaging

- *Portable Doppler US*: Loss of augmentation on calf compression (early). Upon applying the US probe over a patent vein, a continuous sound is heard (*venous hum*). If pressure is applied to the calf, the hum gets accentuated (*augmentation*). If there is thrombosis in the popliteal or femoral veins, accentuation is absent. US is insensitive in calf vein thrombosis.
- *Duplex US*: This method combines both Doppler US flow with US imaging. It can see thrombus and ↑ loss of venous flow (late) or loss of augmentation. Its accuracy approaches that of venography, but is safer & simpler.
- *Ascending Venography* (Figure 3) *invasive*, but the most accurate. A tourniquet is applied just above the malleoli to block the superficial veins. One of the veins of the dorsum of the foot is cannulated and the dye is injected. DVT is diagnosed by failure of visualization of the deep veins or by the presence of a filling defect. Venography although very accurate for the diagnosis of DVT, is invasive and has been almost *replaced* by Duplex scanning.
- ¹²⁵I-fibrinogen Uptake in a thrombus depends upon the incorporation of isotope-labeled human serum fibrinogen into the developing thrombus. It is very accurate in thrombosis of the calf & lower thigh but *not* applicable clinically.
- *Pulmonary scan*: It is done if PE is suspected.
- *Abdominal US or CT scan* if internal malignancy is suspected.
- *CT chest*: In SVC obstruction to R/O mediastinal syndrome.



Figure 3. Ascending venography showing filling defect in the left iliac vein and lower IVC

Differential Diagnosis of DVT

1. *Contusion of calf muscles*: Hx of trauma that may be trivial, ecchymosis and *crescent sign* appearance (Figure 4)
2. *Rupture of plantaris tendon*: Both previous conditions occur during exercise and → swollen painful calf simulating DVT. Dx is by Duplex.
3. *Calf hematoma*: May occur after tibial plateau fissure fracture.
4. *Ruptured Baker's cyst*.
5. *Lymphedema* due to lymphatic obstruction → chronic & usually non-pitting swelling



Figure 4. Contusion of the calf muscles

6. *Cellulitis*: systemic symptoms and signs of inflammation (hot and red).
7. *Calf abscess*: Fever, manifestation of inflammation, ↑ WBCs and pus.
8. *Superficial thrombophlebitis* (hotness, pain and cord-like superficial veins)
9. *Myositis*, especially in diabetic patients
10. *Rupture of superficial femoral artery (SFA) aneurysm* (rare).

Complications of DVT

1. Pulmonary embolism (PE).
2. CDVI and PPS
3. 2^{ry} varicose veins

Management of DVT

Prevention of post-operative DVT (Avoiding Venous Stasis)

1. Early ambulation after operations.
2. Active leg exercises while in bed.
3. Adequate post-operative hydration.
4. Elastic stocking support, especially in the elderly.
5. Intra-operative intermittent pneumatic external calf compression.
6. Prophylactic anti-coagulants

Indications

- High-risk patients for development of DVT.
- History of DVT or PE.
- Major surgery, particularly cancer operations.
- Females on contraceptive pills - Elderly patients - Obesity.

Methods

- Low-dose heparin (Mini-Heparin), 5000 IU, SC 2h before operation & then every 12h until the patient is ambulant (5-7 d).
- Low molecular weight heparin (LMWH), once daily (has lower risk of bleeding).

Treatment of DVT

The objectives of treatment are

- Prevention of formation of new thrombi.
- Prevention of PE.
- To minimize venous valves damage.

A. Bed rest & elevation of the LL

- The patient should be confined to bed with the feet elevated 45 degrees above the level of the heart. Elevation reduces edema & pain, and ↑ venous return (VR).
- Thrombi usually take 7-10 days to become adherent to the vein wall. Graduated ambulation with elastic support is then allowed & continued for 3-6 m until re-canalization & collateralization develop.

B. Anti-coagulant therapy

a. Heparin

Action: It binds to anti-thrombin III, ↑ its activity 1000 times + anti-platelets effect.

Complication of heparin

- Hemorrhage due to over dosage: epistaxis, bleeding gums, hematuria, or GI bleeding. Treatment = IV Protamine sulphate (1mg/100IU of heparin).
- Osteoporosis & alopecia in long-term therapy.
- Heparin-induced thrombocytopenia (**HIT syndrome**): either acute or delayed, dose dependent or independent → ↓ platelets number and ↑ adhesiveness and thrombosis.

Administration

- Heparin is given until disappearance of pain and tenderness of the calf (7-10 d).
- It is given by one of the following methods:
 - ✓ *Continuous IV infusion:* It requires hospitalization and a pump, which controls the dose/h. A bolus dose. of 10000 IU is given IV and then a Dextrose 5% drip containing heparin is infused at a rate of 750-2000 IU/hour for an adult. The dose is monitored by measuring the activated partial thromboplastin time (APTT), which should be 2-2.5 times the control value.
 - ✓ *Bolus therapy:* 5000 IU, IV every 4-6 hours, the APTT is performed one hour before every injection until the dose is adjusted.
 - ✓ LMWH heparin is given SC in a dose of 1 mg/kg/12h. The dose does *not* require blood tests for adjustment.

b. Oral anti-coagulants (Coumarin derivatives-Warfin).

Administration

- Basal level of prothrombin time (PT) and concentration are estimated prior to start of warfarin (5 mg/dl).
- Discontinue heparin after 3 d of overlap treatment.
- Five days after the start of warfarin, the PT and concentration (PC) are measured and the dose is adjusted to reach the therapeutic level of PC 30-40%.
- The international normalized ratio (INR) should be prolonged to between 2-3 times the control values.
- Oral anti-coagulants are given till the time needed for re-canalization & collateralization as evidenced by Duplex examination. It depends on the level of

DVT i.e. if it is:

- ✓ For calf DVT around 3 months
- ✓ Femoral DVT around 6-9 months
- ✓ Iliac DVT around 12-18 months
- ✓ Those with recurrent DVT and congenital coagulation defect by prophylactic Warfarin for ever

Complications: Bleeding is treated by lowering the dose & giving 10-20 mg vitamin K

Contraindications to anti-coagulant therapy

1. *Absolute:* (1) Trauma to or recent operation on the brain or spinal cord & (2) hemorrhagic diathesis.
2. *Relative:* (1) Major visceral injury, (2) major acute fractures, (3) uncontrolled hypertension, (4) History of cerebral hemorrhage and (5) active peptic ulcer.

C. Thrombolytic Therapy

- *Fibrinolytic activators* dissolve fresh thrombi and produce rapid clearance of the occluded veins and may preserve the competence and function of venous valves better than anti-coagulant therapy. They, thus, limit or prevent the development of the post-phlebitic syndrome.
- Streptokinase, Urokinase and Tissue Plasminogen Activator (TPA) are effective only if given in the first 3 d of thrombosis.
- *Disadvantage:* Very expensive
- *Complications:* Allergic reactions - severe bleeding

D. Operative Treatment

1. *Venous thrombectomy:*

- Using Fogarty venous catheter with the use of a protective occlusive balloon proximal to guard against PE.
- *Indications:* In severe cases e.g. PCD with impending venous gangrene.

2. *Insertion of vena cava filter* (Figure 5) is *indicated* in:

- DVT or documented PE in a patient who has a contra-indication to anti-coagulation e.g. previous intra-cerebral hemorrhage or history of bleeding peptic ulcer.
- Recurrent pulmonary embolism despite adequate anticoagulation.
- Complications of anti-coagulation that forced therapy to be discontinued.
- Chronic PE in patients with pulmonary hypertension and cor-pulmonale.
- Patient with severely damaged lung such as interstitial pulmonary fibrosis, or previous pneumonectomy.
- Poly-traumatic patients where liability of the use of anti-coagulants is *not* feasible while the patient is at high risk of developing DVT due to hyper-coagulable state as a response to trauma. A "temporary or retrievable filter" is used in such cases.

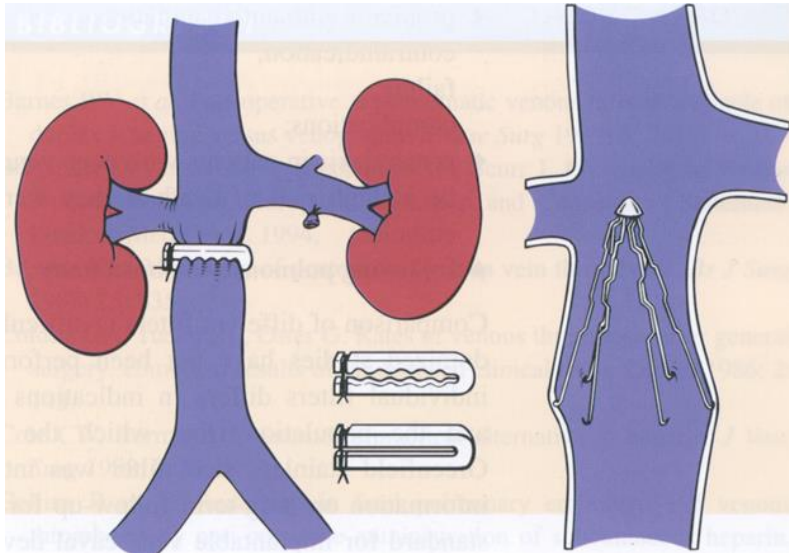


Figure 5. Different methods of interrupting IVC; clipping (to the left) and IVC filter (to the right)

Axillary-Subclavian DVT

Etiology

1. *Primary (spontaneous):* This is usually preceded by unusual muscle activity of the UL (sports, or occupational) due to compression of the vein in the costo-clavicular space.
2. *Secondary* due to central venous catheters, metastatic tumors in the axilla, or IV chemotherapy or parenteral hyperalimentation given via UL veins.

Clinical Features

- Pain and swelling of the entire UL.
- Fullness of the infra-clavicular fossa.
- Prominent venous collaterals over the shoulder & anterior chest wall.
- The fingers may be cyanosed.

Treatment

- Treatment is similar to that of lower limb DVT.

VARICOSE VEINS (VV) OF THE LOWER LIMB

Definition

- The superficial veins of the LL are dilated, elongated and tortuous.

Types

1. Primary Varicose Veins (1^{ry} VVs)

- The cause is not known.
- Theories of etiology
 - a. Inherited weakness of the vein wall, producing venous dilatation and 2^{ry} valvular incompetence.
 - b. The congenital valvular incompetence theory.
 - c. Aggravating factors: prolonged standing, marked obesity, multiparity, intake of estrogens, female gender and constricting clothes.

2. Secondary Varicose Veins (2^{ry} VVs)

- Secondary VVs are not a disease of its own. It is actually one of the manifestations of a disorder of the deep venous system, usually following DVT, hence the name *post-phlebitic syndrome*.
- These are varicose veins that develop 2^{ry} to one of the following:
 - a. DVT: Re-canalization of the thrombosed deep veins leaves the valves of the perforating veins incompetent → reflux of blood to superficial veins which progressively dilate
 - b. AVF: Varicose veins are prominent with traumatic or congenital AVF (**Klippel-Trenaunay syndrome**) (Figure 6) Here, the arterial pressure is transmitted to the deep system → valvular incompetence and reflux of blood to superficial veins → dilate and become varicose.
 - c. Compression of pelvic veins by pelvic tumors or pregnancy. The pressure in the deep venous system of the LLs ↑ → reflux of blood to the superficial system & valvular incompetence.



Figure 6. Klippel-Trenaunay syndrome

Grades of Varicose Veins

1. Dilated venules (hyphen webs or telangiectasia) extend outward from a single feeder vessel (star veins).
2. Dilated tributaries of stem veins (reticular vein)
3. Dilated saphenous veins (stem or tubular veins)

Clinical Features

Symptoms

1. *Cosmetic disfigurement* (dilated veins) (Figure 7).
2. *Aching & discomfort* of the limb and *restless leg*, usually described as dull, heavy, bursting sensation with sense of hotness, usually at the end of the day, or on prolonged standing & is relieved by elevating the limbs.
3. *Night cramps* due to changes in the caliber of calf & soleal sinusoids.
4. *Mild swelling* occurs at the end of the day, particularly with 2^{ry} VVs.
5. *Pigmentation*. This is more in 2^{ry} VVs due to oozing of blood in the SC tissue & deposition of hemosiderin.
6. *Pruritis* may be associated with venous eczema in the ulcer-bearing area (lower 1/3 medial aspect of the leg).
7. *Ulceration* is more with 2^{ry} VVs



Figure 7. Extensive varicose veins of the left lower limb

8. Symptoms in relation to the cause as
 - Pregnancy, DVT, or pelvic obstruction.
 - Other mesenchymal defects: piles, varicocele or hernia.
 - +ve family history of other mesenchymal defects.
9. *Complications*: Superficial thrombophlebitis and venous bleeding after minute trauma.

General Examination

- The patient should be examined while *standing* & should be *exposed* up to the umbilicus.
- Look for predisposing cause as:
 1. Pregnancy, pelvic tumors (PR and PV may be mandatory).
 2. Testicular tumor: para-aortic LN → compression of IVC
 3. Varicocele, piles, or hernia.
 4. Dilated veins on abdomen in 2^{ry} VVs and direction of filling “*Harvey’s test*”

Local Examination

Inspection (Describe the varicosity & skin changes)

- Identify dilated, elongated & tortuous veins:
 - *Site*: according to the involved veins.
 - *Shape*: may be tubular, saccular or serpentine.
 - *Distribution*: according to the type of varicose veins:
 - In 1^{ry} VVs, varicosities may belong to the LSV (Figure 8) or SSV.
 - In 2^{ry} VVs, varicosities are arranged haphazardly
- Effect of elevation of leg: It → emptying of veins if 1^{ry}.
- Varicosities of anterior abdominal wall, vulva and crossing the supra-pubic regions if 2^{ry}.
- Ulcer, or scar of a previous ulcer and eczematization.
- Scar of previous operation.
- Saphena varix, or blow-out.
- Venous stars (mainly on the dorsum of foot and medial aspect of the thigh)
- Edema, especially at the ankle level
- Brown pigmentation & liposclerosis



Figure 8. Varicose veins of the left LL along distribution of the LSV

Palpation and Percussion

- For identification of the course of veins in *obese* patients
- For detection of lipo-dermatoseclerosis
- *Squeeze venous stars*: They fade away and refill.
- Site of defect in the DF (*Fegan’s test*) after marking the sites of veins & emptying them by leg elevation
- *Cruvieller’s test*: Thrill is detected at the SFJ; sometimes, it gives false +ve results.
- *Percussion* to know the tributary belong to main stem or not (i.e. percuss from downward upwards it only locate the vein but not indicate incompetence)
- *Tapping of vein*, from above downward: if +ve indicates incompetent valves in the intervening segment (Schwartz’s test). While the patient is standing, choose an evident

segment of the varicosed vein. Do sudden percussion by the fingers over the upper part of the vein while palpating the lower part.

- Thrill over veins if AVF

Auscultation for bruit in AVF

Special Tests (Tourniquet test, Perthe's test)

Trendlenburg's Test (Single tourniquet)

Aim	Detection of sapheno-femoral junction (SFJ) incompetence.
Method	- The patient lies supine & raises his leg to empty the veins. - A venous tourniquet is applied just below the saphenous opening. - The patient is then asked to stand up. - Observe the filling of the vein while the tourniquet in place and after removal of the tourniquet.
Results	1. If the sapheno-femoral valve is normal, the veins will fill slowly from below upwards. 2. If the sapheno-femoral valve is incompetent, the veins will fill rapidly from above downwards once the tourniquet is released. 3. If the veins fill rapidly while the tourniquet is still in place and filling of veins ↑ on removing the tourniquet, it means incompetent perforating veins in the thigh or leg and SFJ incompetent. 4. If there is rapid refilling of vein while tourniquet is still in place and nothing more change after removal of tourniquet, it means incompetent perforating veins in the thigh or leg with competent SFJ, which is not quite common. In both 3 & 4, <i>multiple tourniquet test</i> is done to identify the site of the incompetent perforators.

Multiple Tourniquet Test

Aim	Localizing incompetent perforating veins when Trendlenburg's test shows rapid filling of varicosities while the SFJ is still occluded (tourniquet in place).
Method	The steps are similar to those of the Trendlenburg's test, but with the application of 3 tourniquets; one just below the knee, the 2 nd at the junction of the lower 1/3 of the thigh with the upper 2/3, and the 3 rd tourniquet at the junction of the lower 2/3 of the thigh with the upper 1/3. The patient is then asked to stand up & the tourniquets are released one by one from below upward.
Result	Rapid filling in any segment indicates the incompetency of the valve of the perforator above the released tourniquet. This test is greatly replaced by a more accurate non-invasive duplex examination.

Perthe's Test

Aim	Detection of the patency of the deep venous system.
Method	- The patient lies on his back and the LL is elevated. - Elastic bandage is applied firmly from the toes to the upper 1/3 of the thigh. - The patient is then asked to stand & walk in situ for 5 minutes
Result	If the deep system is blocked, the patient will complain of pain in the leg and inability to continue the exercise. The pain is relieved upon removal of the elastic bandage. This test is greatly <i>replaced</i> by Modified Perthe's test

Modified Perthe's test

Aim	Detection of the patency of the deep system
Method	While the patient is standing, a tourniquet is applied just below the SFJ & the patient is asked to walk quickly in situ for five minutes
Result	- If the VVs <i>disappear</i>, this means that the deep system is patent and competent as the efficient leg muscle pump creates a -ve pressure in the deep veins, thus sucking the blood from the superficial venous system. If the veins become more engorged, then the deep veins are obliterated or incompetent. - There is also <u>severe pain</u> that force patient to stop exercise.

Complications of Varicose Veins (more with 2^{ry} VVs)

- Edema of the LL usually starting at the ankle.
- Itching, dermatitis and eczema due to subcutaneous deposition of hemosiderin
- Brown pigmentation, especially on the medial aspect of lower third of the leg.
- Lipo-dermato-sclerosis: Extravasated fibrinogen → FT formation replacing the soft SC fat.
- Recurrent superficial thrombophlebitis.
- Venous ulceration.
- Combination of the above-mentioned complications + chronic leg pain = "**post-phlebotic syndrome**" as they usually follow previous DVT with incompetent perforating veins.
- Hemorrhage may occur due to rupture of a VV, especially when the overlying skin is thin. It can be initially stopped by elevation and compression bandage. Later treatment is by injection sclerotherapy. It must *not* be stopped by stitches, which may be the start of a new ulcer.
- Malignant transformation (**Marjolin ulcer**).
- Equinus deformity due to toe-walking.

The CEAP Classification System of Venous Insufficiency

Clinical signs	(A) for asymptomatic and (S) for symptomatic
Etiologic classification	Congenital, primary or secondary
Anatomic Distribution	Superficial, deep, perforating, or combination
Pathophysiologic dysfunction	Reflux or obstruction, alone or in combination

Clinical Classification of Chronic Venous Disease

Class 0	No visible or palpable evidence of venous disease
Class 1	Telangiectasia's or small reticular (1-4 mm diameter)
Class 2	Varicose veins > 4 mm in diameter
Class 3	Varicose veins with edema
Class 4	Skin changes (pigmentation, lipo-dermato-sclerosis)
Class 5	Skin changes with venous ulceration
Class 6	Skin changes with active ulceration.

Investigations

Most cases of 1st VVs do not require investigations.

- Routine tests for the operation (CBC & coagulation profile).
- Tumor markers, US or CT if malignancy in case of 2nd VVs, is suspected.
- Specific tests:

A. Non-invasive

- **Air or photoplethysmography:** Of academic interest & measures the changes in limb volume in relation to exercise, standing, or leg elevation. It can test (1) patency of deep venous system, (2) reflux in deep system & (3) reflux in the superficial system.
- **Portable Doppler US:** Assessment of SF reflux & patency of the deep system.
- **Duplex scan:** as portable, but more accurate in (1) detection of the sites of incompetent perforating veins, (2) verifying patency of the deep veins & (3) localizing the site of SPJ.

B. Invasive

- **Venography either:- (accurate but we rarely need it)**
 - o *Ascending* by applying tourniquet over ankle to occlude superficial veins & inject dye in dorsal veins that goes to deep system. If it goes to the superficial system, this means incomplete perforators. It also can detect any congenital anomalies
 - o *Descending venography* by injecting the dye in the femoral vein – detects the degree of reflux in the deep & superficial systems.
- **Varicography** by injecting the dye in the bunch of veins.
- **Measurement of ambulatory venous pressure (obsolete now).** It shows the following changes in pressure while walking according to the following different clinical situations:

Clinical Situation	Walking	Rest
Normal	Rapid ↓	Slow return to normal
Obstruction	Rapid ↑	Slow return to normal
1 st VV veins with competent perforator	↓	Rapid return
incomplete perforator and incomplete deep venous system or obstruction	little or no ↓	Rapid return.

Treatment

Primary Varicose Veins

A. Conservative Treatment

- Minor or spider varicosities are treated by reassurance and graduated elastic stockings.
- *Indications*
 1. CDVI
 2. Old age
 3. Pregnancy
 4. Unfit patient or refusal of operation.
 5. Post-operatively for residual or recurrent tributaries.

B. Injection Sclerotherapy

- *Indications*
 1. Spider veins.
 2. Localized dilated superficial veins.
 3. Recurrent or persistent veins after operation.
- *Principle*
 1. Sclerosants like 5% Ethanolamine Oleate, or
 2. 3% Sodium Tetradecyl Sulphate.
- *New trends in injection*
 1. Foam injection sclerotherapy by mixing air with sclerosing material this help to inject less material with a wider spreading area.
 2. Injection of the main trunk under Duplex control through a special catheter.
 3. Radio-frequency ablation of superficial veins.
- *Technique (Method)*

Injection should be in an empty vein to induce injury of the endothelial layer of the intima. Compression is then applied; the two walls of the vein will adhere together.
- *Complications*
 1. Extravasation of sclerosant material → sloughing of skin (eczema) and poor cosmesis.
 2. Ulceration
 3. DVT (if a large amount of the sclerosant material reaches the deep system undiluted)
 4. Accidental intra-arterial injection
 5. Recurrence.

C. Surgery

1. *Segmental stripping* of a localized area of diseased veins.
2. *Ligation of SFJ (Trendlenburg's) operation:* It is not a good operation as it allows communication between tributaries of the LSV & SSV.
3. *Endothelial stripping:* Injury of endothelium of veins, then application of elastic bandage → adhesion and fibrosis → ↓ post-operative stay and complications.

4. *Stripping of the LSV or SSV*

- Anesthesia: General, spinal or epidural.
- Position of Patient: Supine for LSV and prone for SSV stripping.
- Indication: SFJ or SPJ incompetence
- Preparation: Antiseptic and towel draping (LL Hair shaving at night of operation.).
- Procedure: An incision is made over the LSV (anterior and above medial malleolus) or SSV (posterior and above lateral malleolus) and expose the vein only (dissect it from saphenous nerve in case of LSV or sural nerve in case of SSV). Ligate the vein distally, make a transverse venotomy and pass the stripper to the junction (SFJ or SPJ). At the SFJ, a transverse incision is made in the groin crease over the palpable stripper or just medial to the femoral artery pulsation. At the SPJ, a transverse incision is done in popliteal fossa better after localizing SPJ by Duplex. Ligate the tributaries of the LSV before its junction with the femoral vein. They are 3-5 (superficial circumflex iliac, superficial inferior epigastric and superficial external pudendal veins) and disconnect the SFJ flush with the vein to prevent long stump that may cause DVT. In case of SSV, it lies deep to the DF. Replace the guiding head of stripper and apply the olive head of suitable size and ligate the vein over it and close the upper wound. Any fascial defect should be identified and closed and any dilated veins should be removed (**buncheonectomy**) or injected. Apply *elastic bandage* over the LL and then strip the vein. Close the lower wound. The leg is elevated post-operatively to ↓ hematoma formation. Remove stitches after 2 weeks. Any *residual veins* are injected post-operatively.

5. *Ablation by radiofrequency or LASER*

- Complications
 1. Injury of the femoral vein (repair it and if the LSV had not been removed, leave it to help drainage of the limb)
 2. Injury of the femoral artery.
 3. Hematoma.
 4. Injury to the saphenous nerve → anesthesia in the medial aspect of the leg (so do *not* to strip *below* level of mid-calf)
 5. Post-operative DVT.
 6. Recurrence due to double saphenous or failure to identify the 3 tributaries.

Secondary Varicose Veins

1. *Post phlebitic* (after DVT): Most cases are treated conservatively by elastic stockings. Rarely, varicosities are large enough to require localized excision or stripping.
2. *Varicose veins complicating AVF*: Surgical treatment of the fistula is usually followed by marked regression of varicosities.
3. *VVs during pregnancy*: A complete elastic stocking from the toes up to the groin is applied during pregnancy. After labor, small residual veins are treated by sclerotherapy.

VENOUS ULCERS

Incidence

- Venous ulcers constitute 90% of chronic leg ulcers.

Etiology

1. Venous ulcers usually occur in a post-phlebitic limb due to previous DVT.
2. A much less common cause is arterio-venous fistulas (AVF).
3. Rarely, the ulcer is a complication of 1st VVs, usually transient & heals rapidly.

Pathogenesis of Chronic Deep Venous Insufficiency CDVI (PPS)

- ↑ pressure in the deep system following re-canalization of DVT due to dysfunction of the valves of both, deep & perforating veins + narrowing of a large deep vein.
- As the skin of the lower part of the leg is drained directly by the *ankle perforators*, which drain into the deep system, incompetence of these perforators → high venous pressure in the skin on the medial side of the lower 1/3 of the leg (**ulcer-bearing area**).
- Venous hypertension resulting in;
 1. ↑ hydrostatic venous pressure during standing from the right atrium to the leg (**static**).
 2. **Dynamic** venous hypertension due to incompetent perforator.
- *Both result in:*
 1. Escape of RBCs with deposition of hemosiderin in the lower 1/3 of the leg, especially at the medial aspect leading to itching and weeping dermatitis.
 2. ↑ WBCs mainly T-lymphocytes and macrophages trapping → leucocyte proteolytic enzymes release & free-radical activity resulting in destruction of the skin.
 3. ↑ fibroblastic activity → fibrin deposition around the capillaries with peri-capillary cuff formation resulting in hypoxia to the tissues and itching with trauma and ulceration.
 4. Lipo-dermato-sclerosis due to fibrosis of SC tissues

Clinical Features (Figures 9-14)

- Past history of DVT (*not* present in all cases).
- It is usually situated in the ulcer-bearing area or around the medial malleolus.
- Pain (severe, bursting or tightness, especially on walking “**venous claudication**”)
- Edema, soft pitting at the beginning and later on, firm and refractory to recumbency.
- Skin around the ulcer → eczema, itching marks, induration and dark brown pigmentation.
- Lipo-dermato-sclerosis (tough skin and SC) → inverted **sampan bottle** appearance of the leg
- Secondary VVs may be present.
- The ulcer takes a long time to heal.
- Recurrent ulceration begins by ankle flare then pre-ulcer & lastly ends by an ulcer



Figure 9. Varicose ulcer



Figure 10. Post-phlebotic ulcer



Figure 11. Bilateral 2nd VV ulcer



Figure 12. Post-phlebotic ulcer with pigmentation



Figure 13. Venous ulcer with pigmentation & liposclerosis



Figure 14. Venous ulcer with inverted bottle appearance

- The ulcer has the following characteristics:
 - Painful
 - *Site:* Behind & above medial malleolus (usually).
 - *Margin* of the ulcer is congested and cyanosed.
 - *Edge* is slopping
 - *Shallow*, may penetrate to the DF, but *not* through it.
 - *Floor* is covered by granulation tissue (GT).
 - *Base* is indurated.
- Differences between post-phlebotic varicose ulcers are summarized in the Table below:

Criteria	Post-Phlebotic ulcer	Varicose ulcer
<i>Cause</i>	Old DVT	Primary varicose veins
<i>History</i>	Chronic, rarely healed	Recent
<i>Pathology</i>	DVT &/or incompetence	VVs of the LSV
<i>Clinical Picture</i>	- Large ulcer above medial malleolus - Lipodermatosclerosis - ↑ pigmentation around the ulcer	- Small ulcer distal to LSV - Limited skin pathology in the lower 1/3 of the leg
<i>Duplex</i>	- DVT &/or incompetent deep system - Incompetent Perforators	- Normal deep vein.
<i>Treatment</i>	- Conservative - Perforator ligation	- Sapheno-femoral disconnection & stripping

Investigation

- Similar to those of primary varicose veins

Differential Diagnosis (DD)

DD According to the Presenting Condition

- *Pigmentation: Kaposi's sarcoma.*
- *Venous claudication: Arterial insufficiency.*
- *Edema: Swollen limb, especially due to lymphedema.*
- *Ulcer: Other causes of chronic leg ulcer*

DD of Chronic Leg Ulcer

Vascular Ulcer

Arterial

- Ischemic ulcer due to all causes of ischemia.
- It occurs at pressure points, very painful & deep (Figure 15), resists R/ + C/P of ischemia.

Venous

- *1^{ry} venous ulcer:* Associated with LSV 1^{ry} VV, superficial, small, painless & usually near the dilated veins + clinical picture of 1^{ry} VVs.
- *2^{ry} venous ulcer* Associated with incompetent perforators (1^{ry} CDVI or 2^{ry} to PPS), may penetrate to DF and tendons but *not* through them. Associated with burning sensation, usually behind and above medial malleolus (*Cocket's* perforators) + C/P of CDVI.

Lymphatic

- Lymphedema due to its multiple causes.
- Clinical picture of lymphedema + ulcer irregular in shape and location ± lymphorrhea.

Traumatic Ulcer

- History of trauma by burn, radiation, decubitus.
- The affect skin is stretched over the bone (*shin of tibia*).
- May be infected where the floor is covered by pus, or healing ulcer with sloping edge.

Inflammatory Ulcer

- *Tuberculosis (TB):* It occurs in, either 1^{ry} TB over the calf muscle as a nodule that ulcerates later, or 2^{ry} TB of the skin (*lupus vulgaris*) over a deeply-seated source of infection (LN, TB bone). The ulcer is small, multiple, painful, with undermined edges, bluish irregular *margin* and a *floor* covered with unhealthy GT. The *LN*s are enlarged, but *not* tender.
- *Syphilis (\$).*
 - 1^{ry} \$: Syphilitic ulcer on the genitalia, tongue & tonsils.
 - 2^{ry} \$: On mucous membranes as snake-tract ulcers with irregular margin.
 - 3rd \$: *Gummatous* ulcer: Dusky red, painless nodules that coalesce and ulcerate. The ulcer has a serpiginous margin, punched-out edge, yellow necrotic wash-leather floor, indurated base and heals with FT.



Figure 15. Ischemic (arterial) ulcer

- *Meleney's ulcer*: Septic ulcer, a type of infective gangrene in a toxic patient. It begins by erythematous indurated areas that ulcerates and forms *multiple painful* ulcers. Treat infection + local care of the ulcer.
- *Parasitic ulcer*: It is caused by *Leishmania Tropica*. It starts as a papule that ulcerates & heals by fibrosis. The ulcer is shallow, small & very chronic.

Malignant Ulcer

- *Skin tumor*; Basal cell carcinoma (BCC), SCC (squamous cell carcinoma – Figure 16), malignant melanoma (MM), Kaposi sarcoma, 2^{ty} skin nodules (ulcerated).
- *On top of burn or chronic wound (Marjolin ulcer)* (Figure 17)
- *Sarcoma*
- *Leukemia*.



Figure 16. Malignant ulcer - SCC



Figure 17. Marjolin ulcer (SCC on top of burn)

Neuropathic

- It occurs due to loss of sensation due to any cause as (peripheral neuropathy, spina bifida, Tabes dorsalis, leprosy and DM).
- Over bony prominence (Figure 18)
- Penetrate deeply, may be infected.
- Painless because of neuropathy and impaired sensation in the nearby skin.

Martorell's Ulcer

- Also known as hypertensive ulcer with *no* atherosclerosis.
- Painful + history of hypertension.

Anemia: Hemolytic anemia e.g. thalassemia.

Mercury poisoning

Prolonged treatment with Corticosteroids

Trophic Ulcers: due to impairment of nutrition (ischemia and anesthesia) e.g. bed sores and DM.

Cryopathic Ulcer: Cold injuries.

Iatrogenic: Extravasation during injection sclerotherapy of VVs.

Ulcer Artefacta: Self-inflicted.



Figure 18. Neuropathic ulcer in a patient with DM

Treatment

- Treatment after DVT by wearing elastic stocking and leg elevation at night.

In Case of Ulcer (Bisguard Method)

- Keep the ulcer clean with leg elevation (↓ stasis) and frequent dressing then apply elastic bandage + apply olive oil and Calamin lotion for eczema surrounding the ulcer and Zinc oxide for the ulcer.
- The use of systemic antibiotics is claimed to be ineffective due to presence of peri-capillary cuff that impedes the delivery of antibiotics to ulcer.

New drugs

- Anti-platelets
- Fibrinolytic drug
- Prostaglandins: PGE1
- Pentoxifyllin (Trental) to improve RBCs passage in the capillaries.
- Free-radical scavenger

Surgery

- Indicated in resistant chronic venous ulcers and in recurrent cases. Venography or Duplex is a pre-requisite for the choice of the proper operation.
- For ulcer and superficial venous system
 - For incomplete perforators : **Cockett's operation** by a longitudinal incision behind and above the medial malleolus and *sub-facial ligation* of incompetent perforators. It can be done *endoscopically*. *Complication*: Bad wound healing due to lipodermatosclerosis.
 - **Linton's operation**: posterior incisions for lateral, medial & posterior perforators.
 - **SEPS: Endoscopic Subfascial Perforator Surgery ligation** is now gaining popularity as it avoids the complications of an incision in diseased skin in the lower 1/3 of the leg near the ulcer.
 - *Large resistant ulcer needs skin graft or flap* (but it is not good due to poor blood supply and exudation due to venous hypertension).
- *Venous reconstructive surgery by*:
 - Valve repair
 - Venous system transfer (from the LSV with its valves to the deep system)
 - **Palma's operation**: Mobilization of the distal end of the LSV of the sound LL supra-pubically to drain the other limb below the site of obstruction.

The syndrome of Chronic Venous Insufficiency (CVI)

Definition

It is a condition characterized by venous congestion in the LL due to impaired venous outflow. It is clinically expressed by LL heaviness and skin signs if venous congestion.

Causes

1. Superficial varicose veins (1^{ry} or 2^{ry}).
2. Chronic deep venous obstruction.
3. Post-phlebitic syndrome.
4. Primary deep veins or perforating vein valvular incompetence.
5. Chronic pelvic or abdominal venous compression (e.g. pregnancy, masses, etc.)